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Howe et al.

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(45) **Date of Patent:** ***Aug. 26, 2025**

(54) **RADIO FREQUENCY COMMUNICATIONS
USING LASER OPTICAL BREAKDOWNS**

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(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 453 days.

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This patent is subject to a terminal dis-
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Primary Examiner — Pranesh K Barua

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(51) **Int. Cl.**
H04B 10/50 (2013.01)
H04B 10/11 (2013.01)
H04B 10/54 (2013.01)

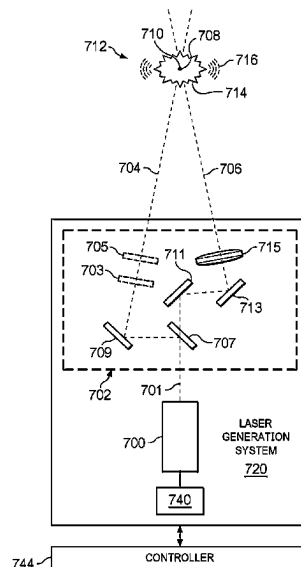
(57) **ABSTRACT**

A method, apparatus, and system for transmitting data. A
radio frequency communications system comprising a com-
puter system and a communications manager in the com-
puter system. The communications manager is configured to
identify data for transmission using radio frequency noise
signals or other electromagnetic spectrum. The communi-
cations manager is configured to control an emission of laser
beams at a set of optical breakdown points to cause optical
breakdowns that generate radio frequency noise signals
encoding the data.

(52) **U.S. Cl.**
CPC **H04B 10/503** (2013.01); **H04B 10/11**
(2013.01); **H04B 10/541** (2013.01); **H04B**
2210/006 (2013.01)

(58) **Field of Classification Search**
CPC H04B 10/503; H04B 10/11; H04B 10/541;
H04B 2210/006
See application file for complete search history.

28 Claims, 25 Drawing Sheets



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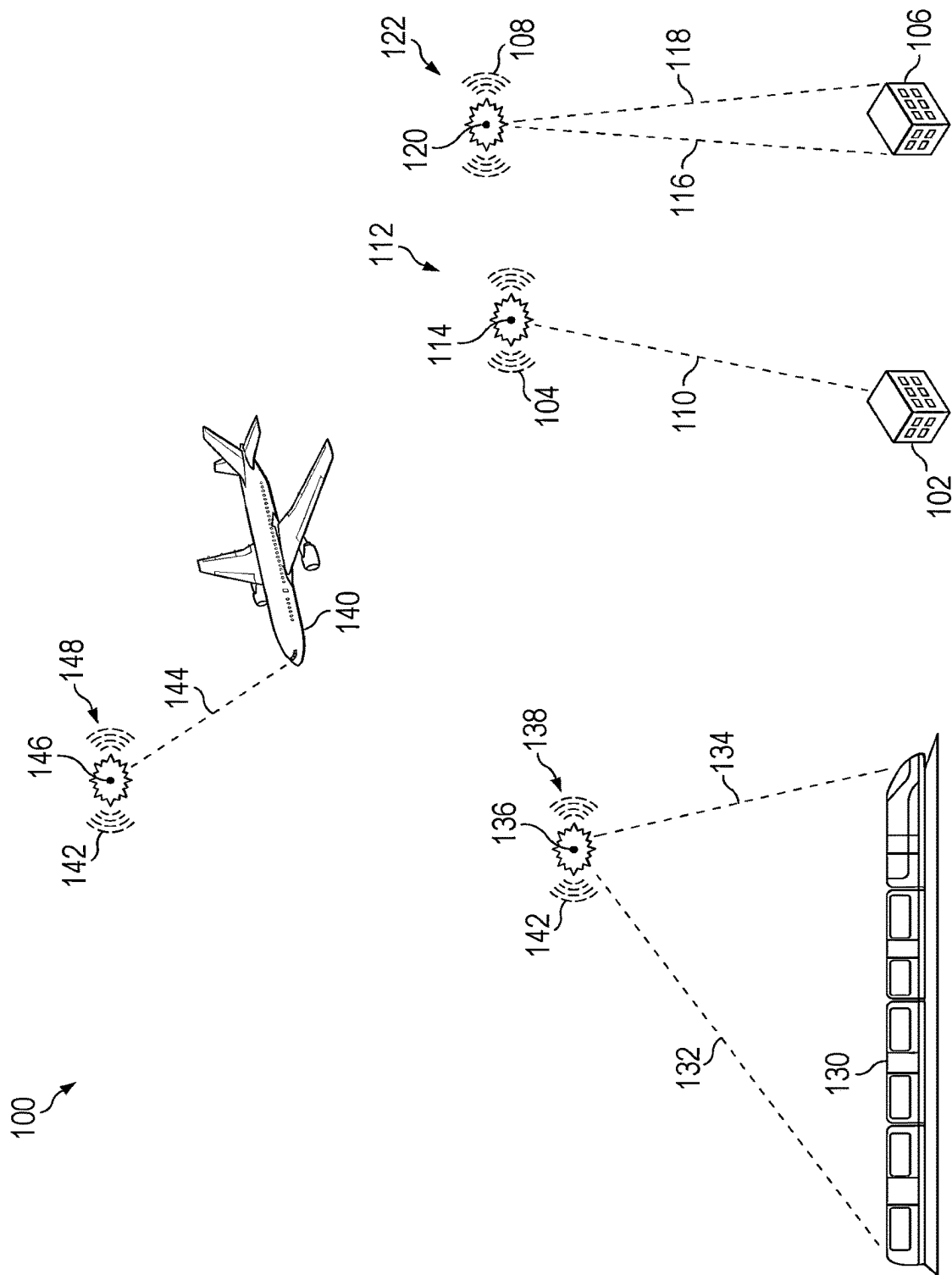


FIG. 1A

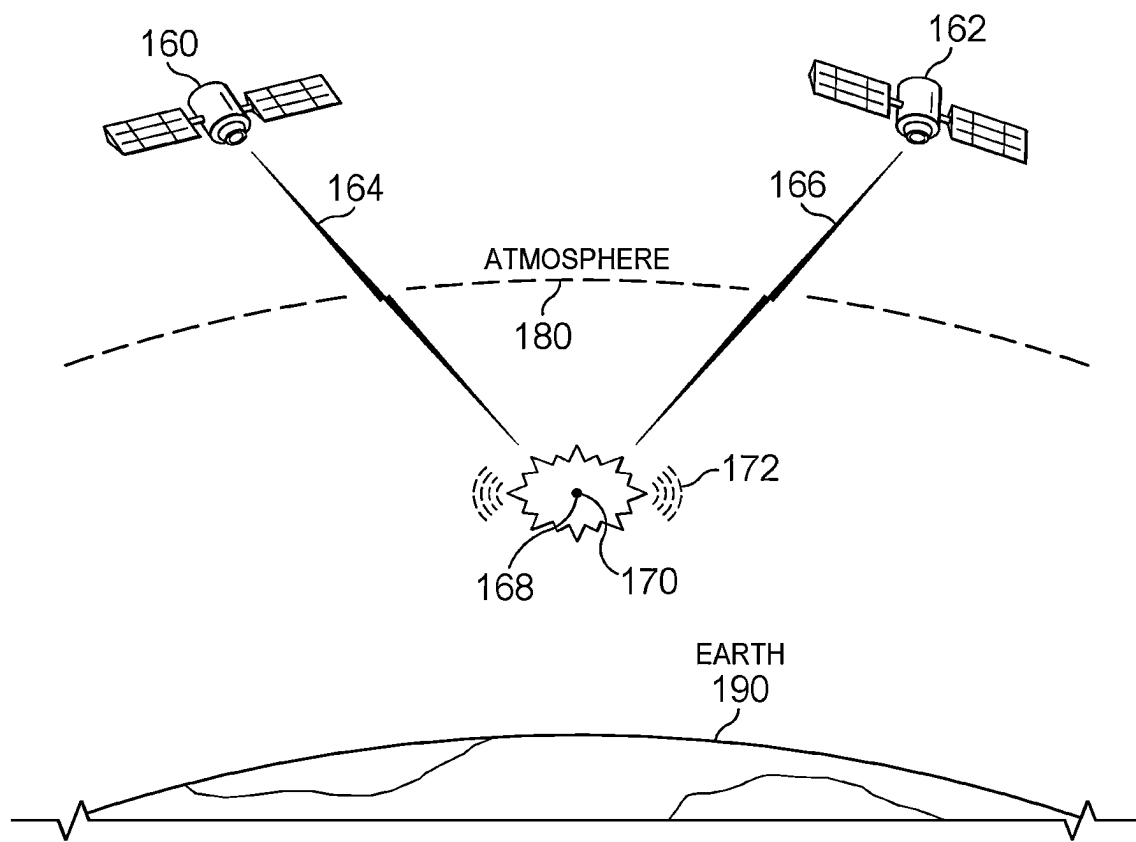


FIG. 1B

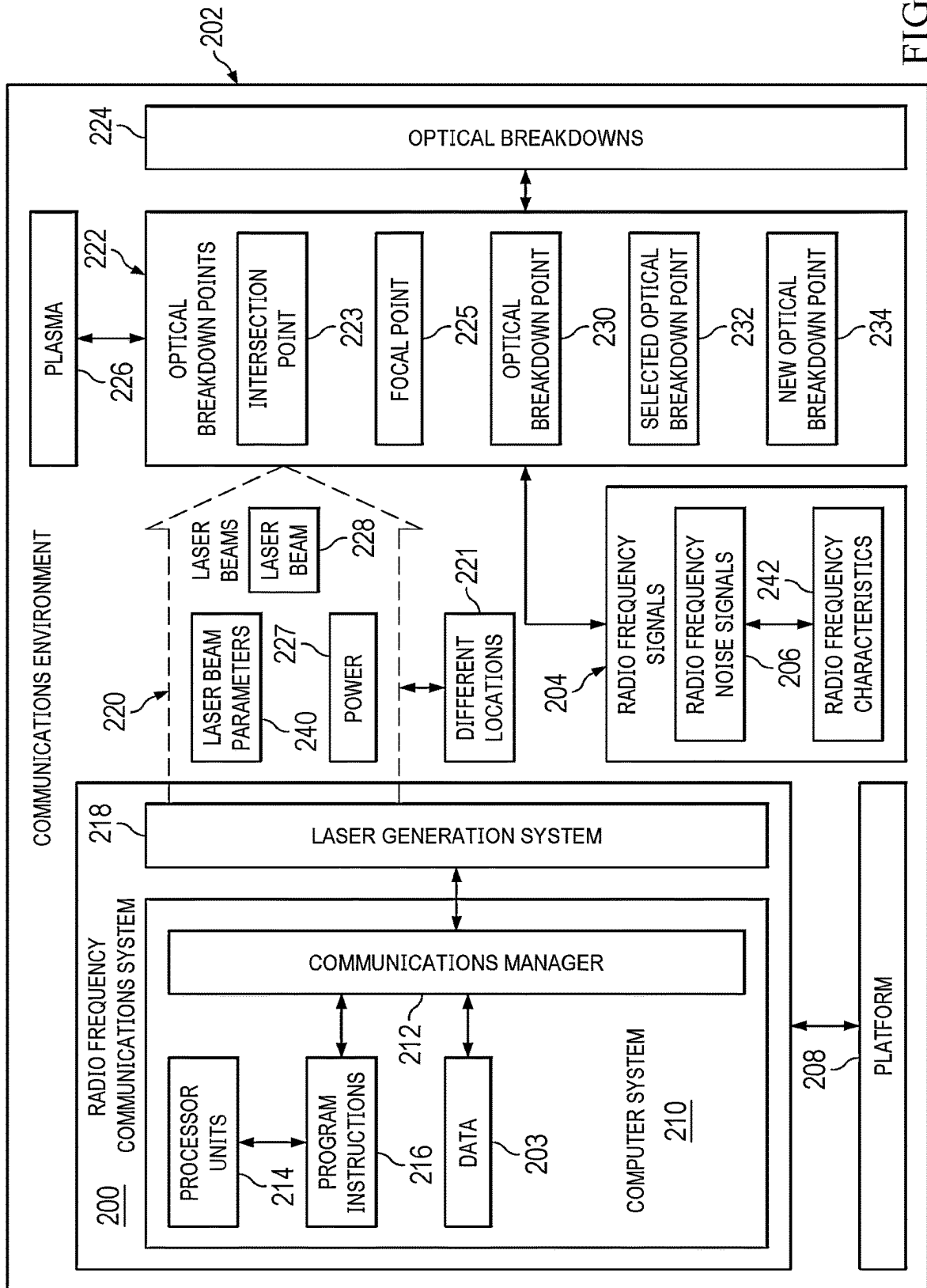


FIG. 2

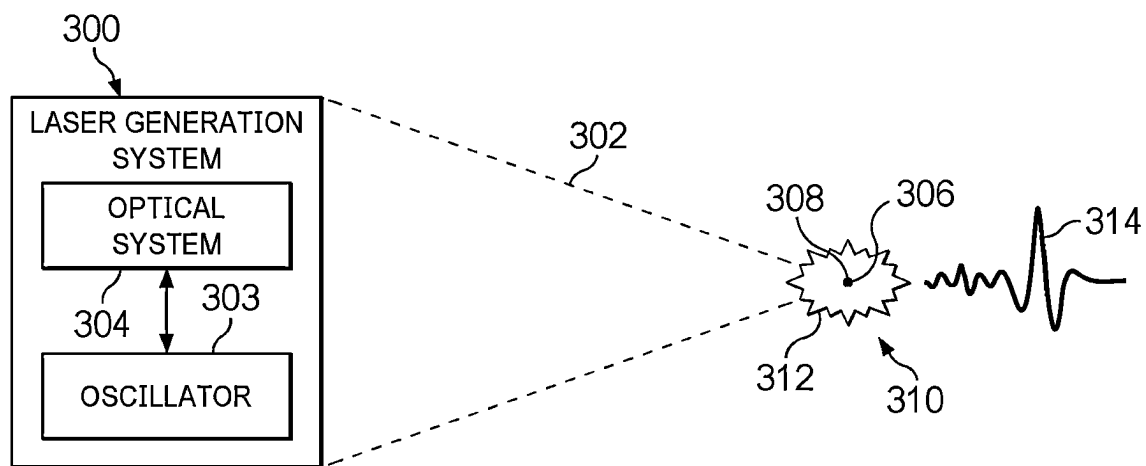


FIG. 3

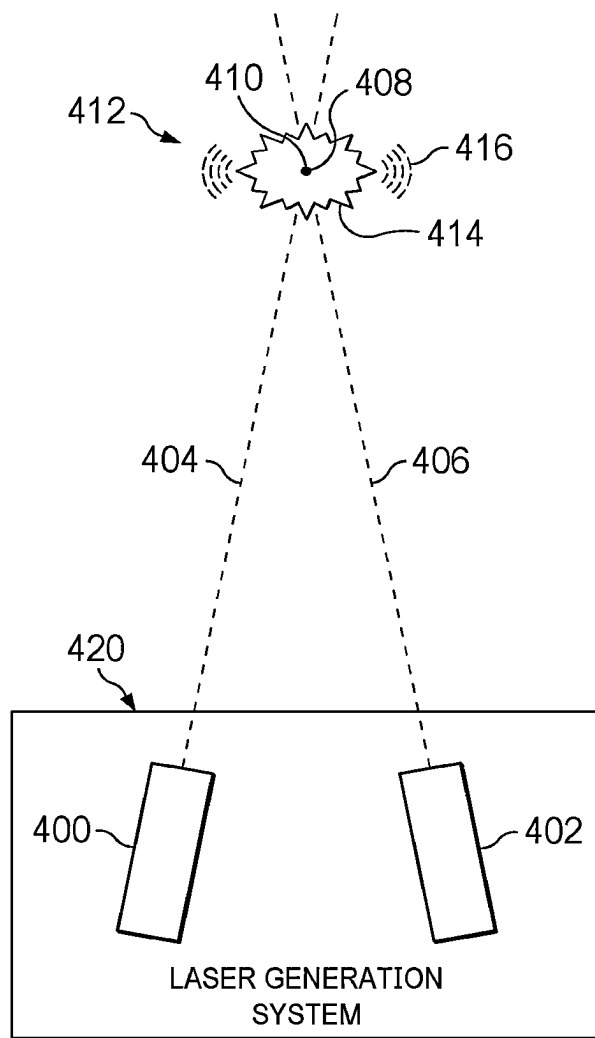


FIG. 4

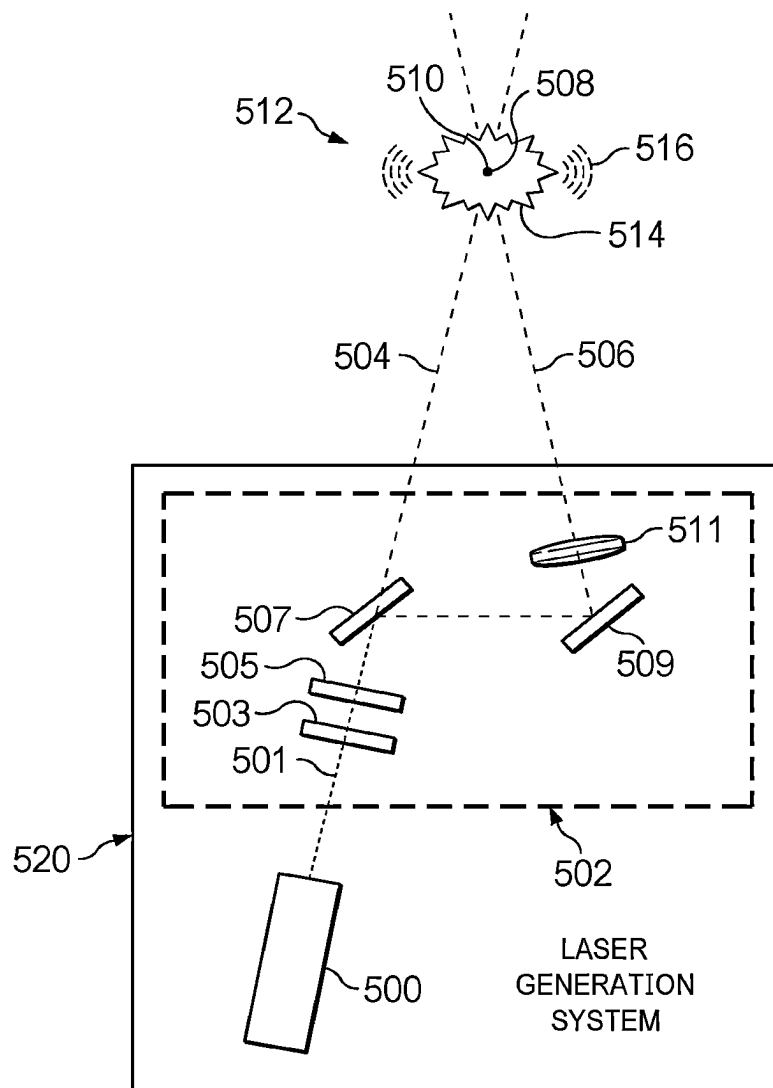


FIG. 5

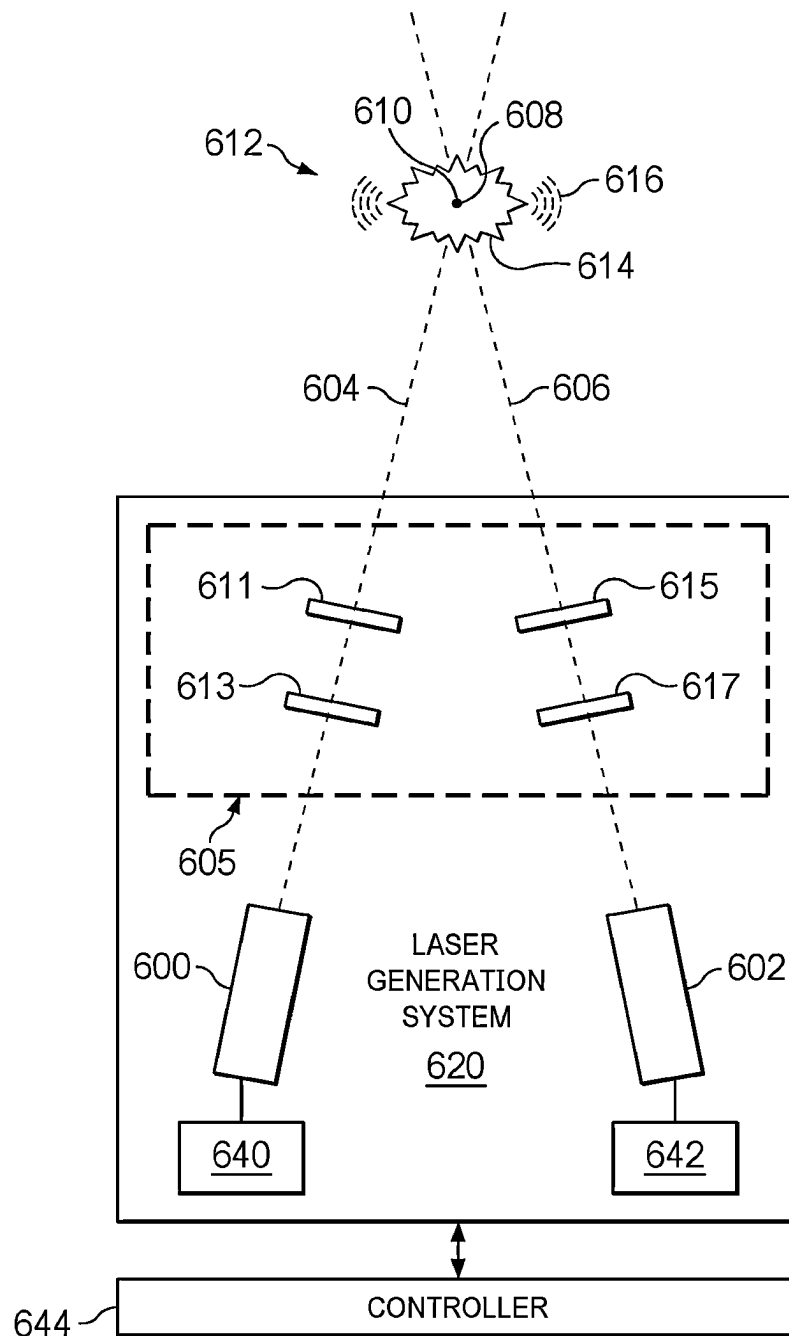


FIG. 6

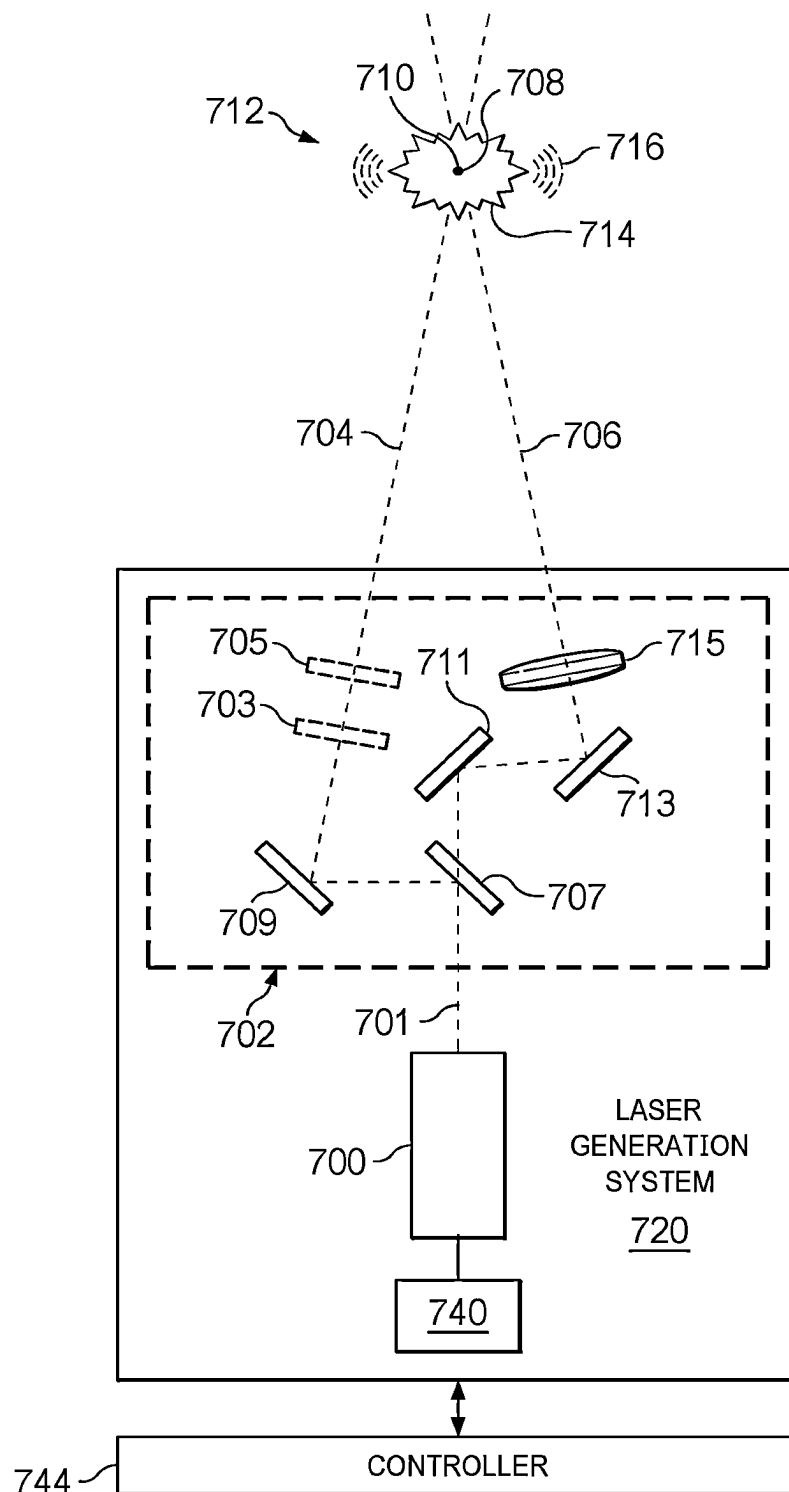


FIG. 7

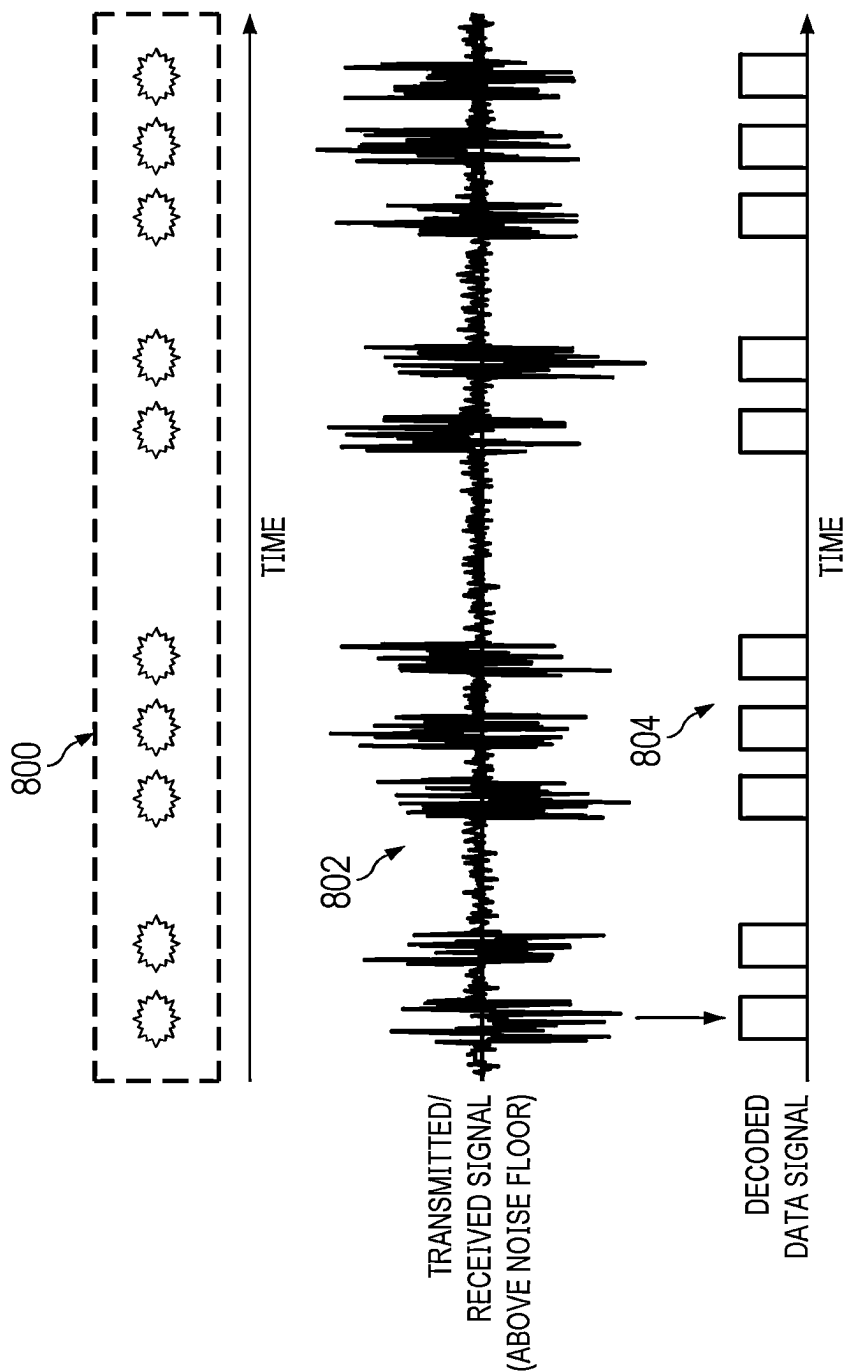


FIG. 8

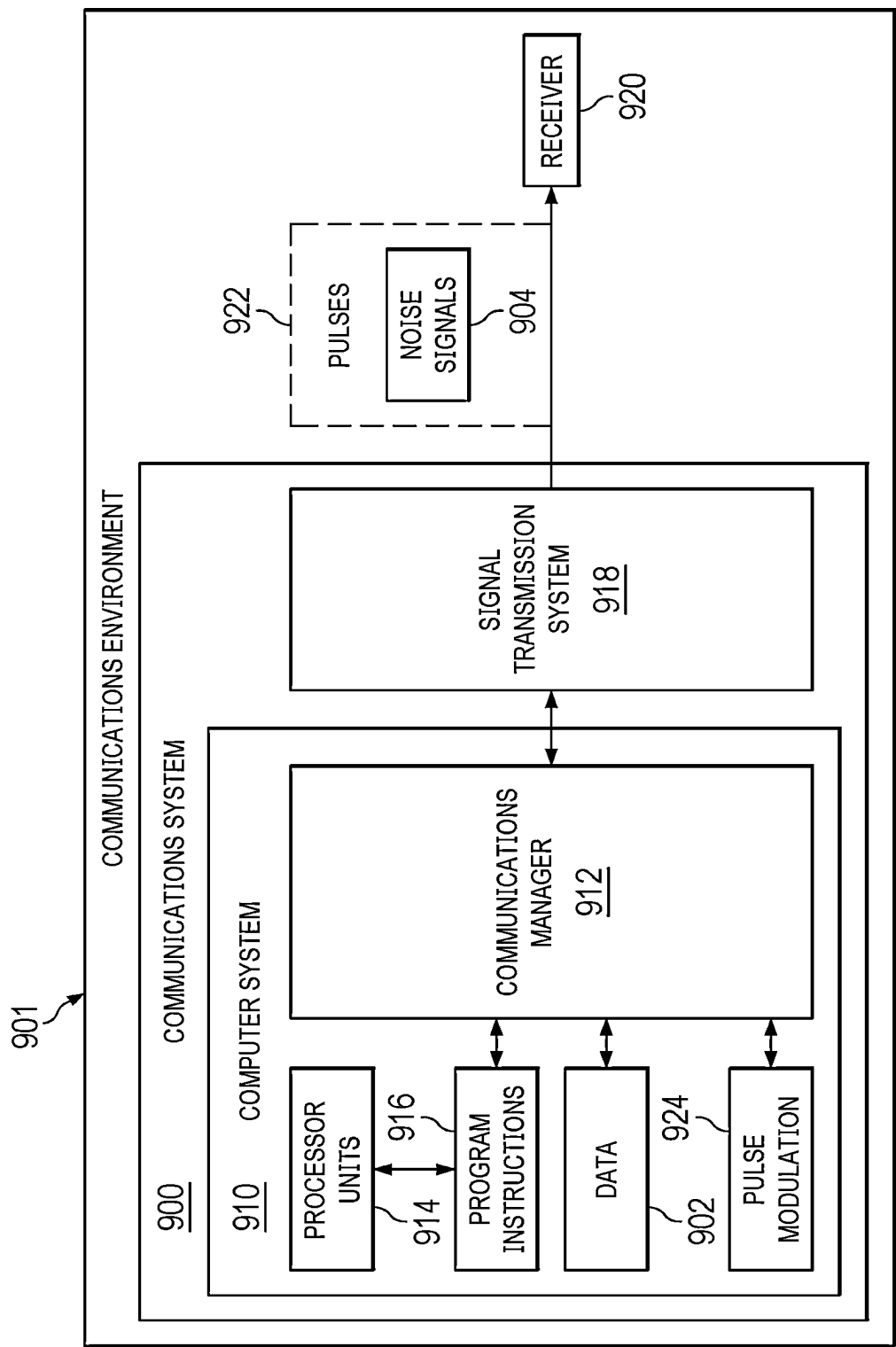


FIG. 9

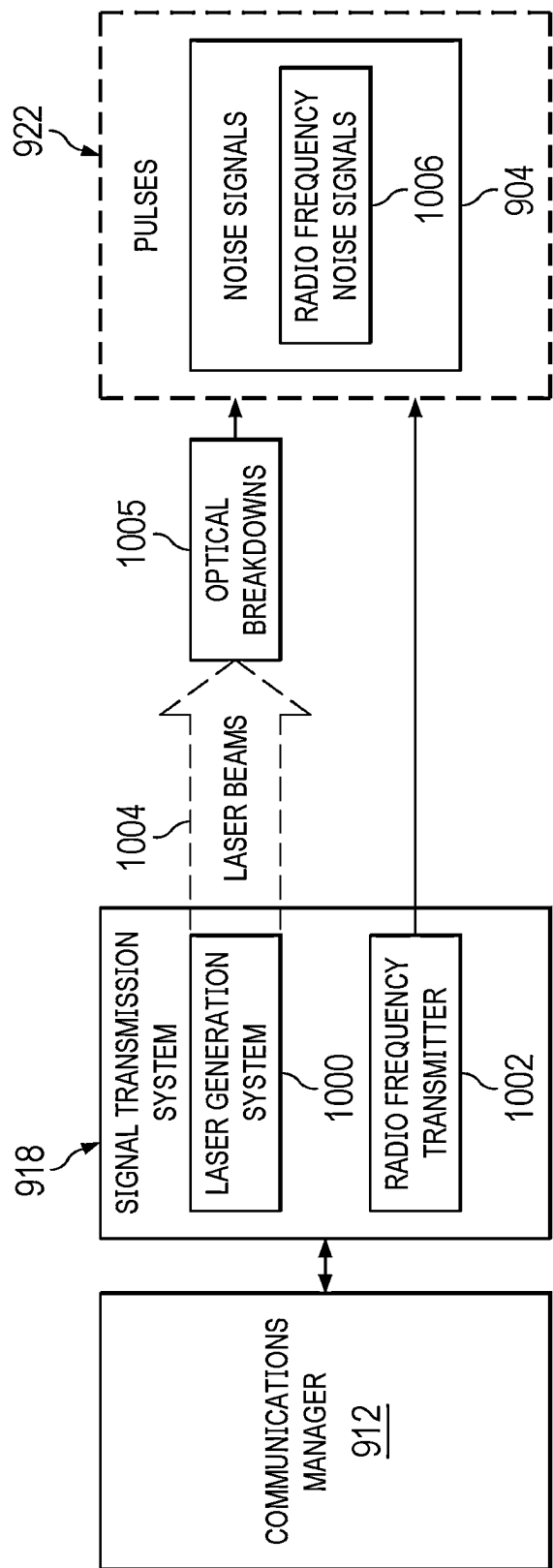


FIG. 10

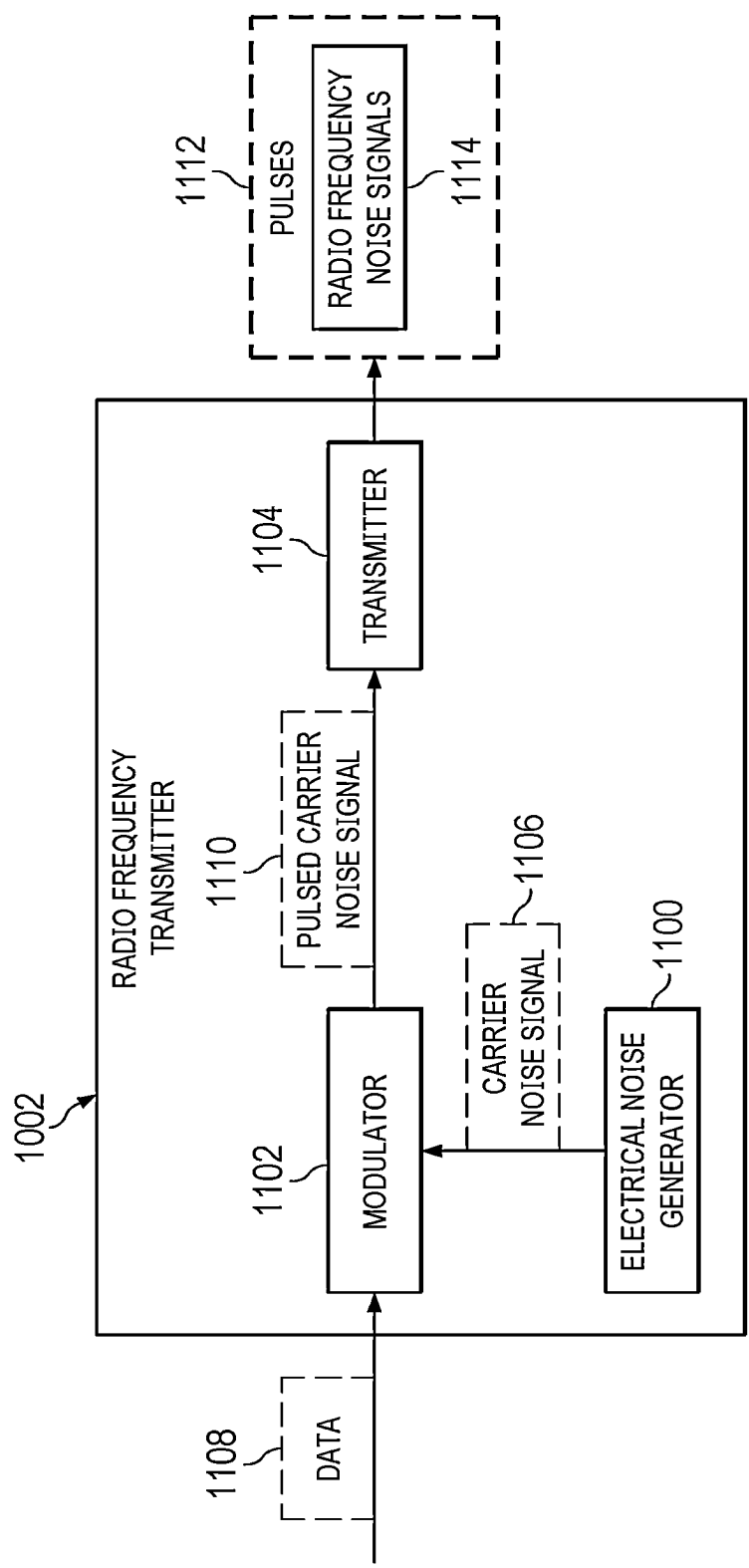


FIG. 11

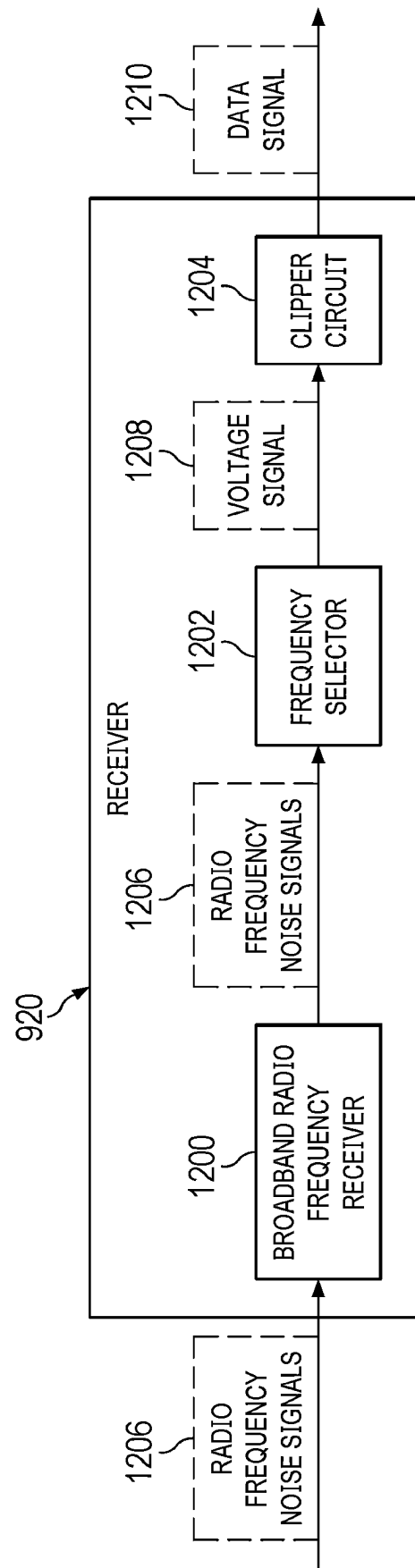


FIG. 12

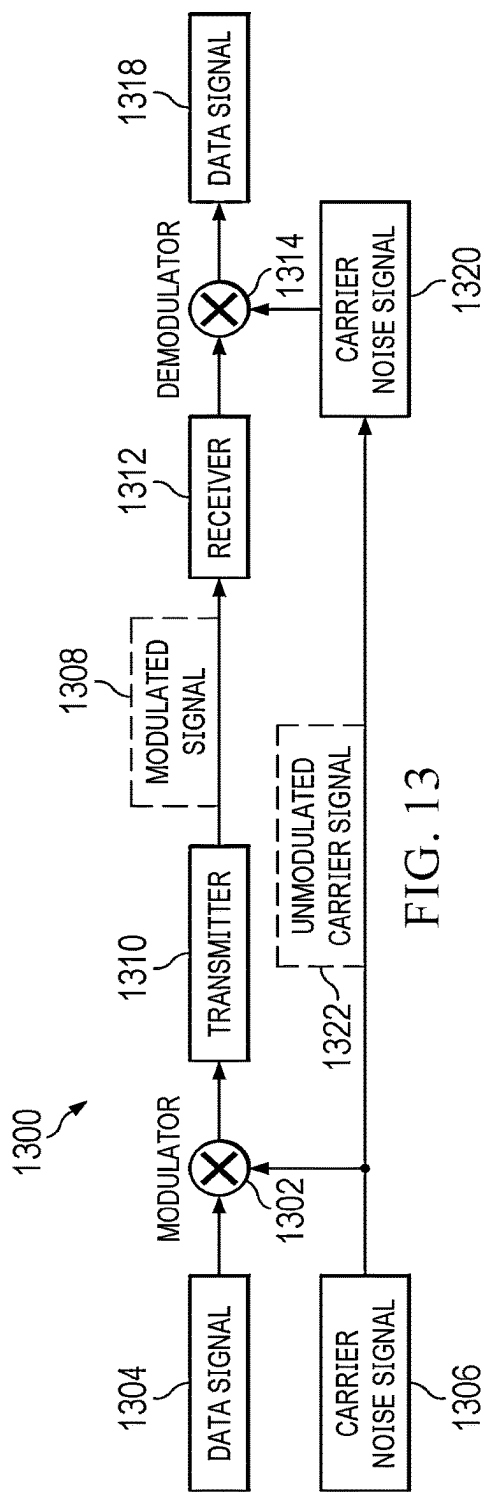


FIG. 13

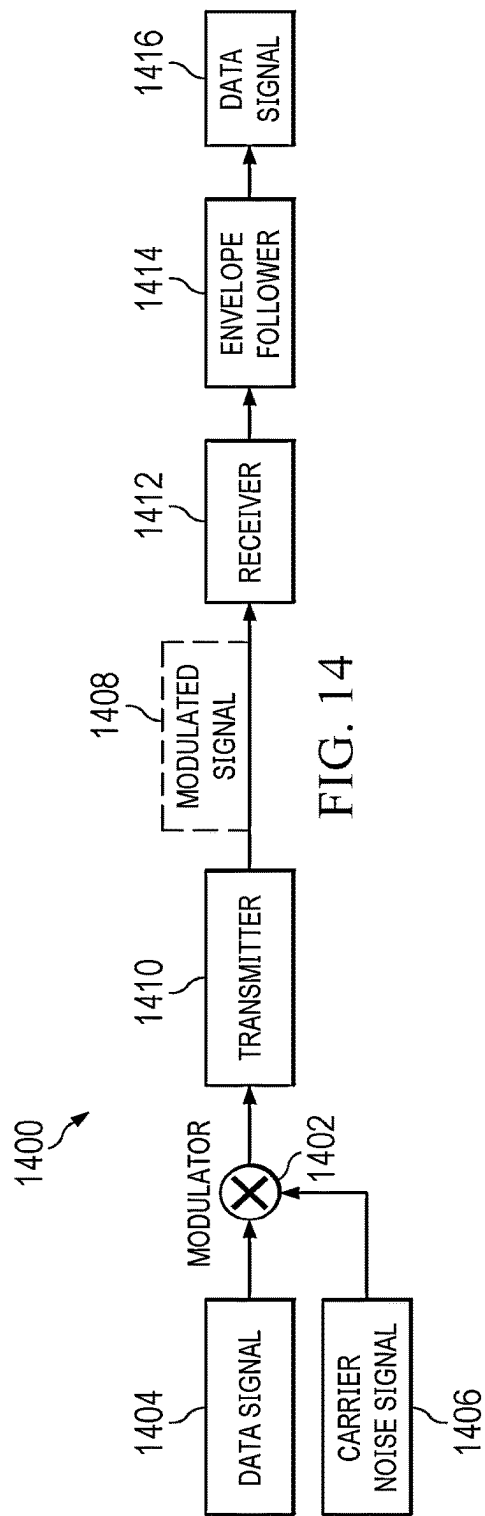


FIG. 14

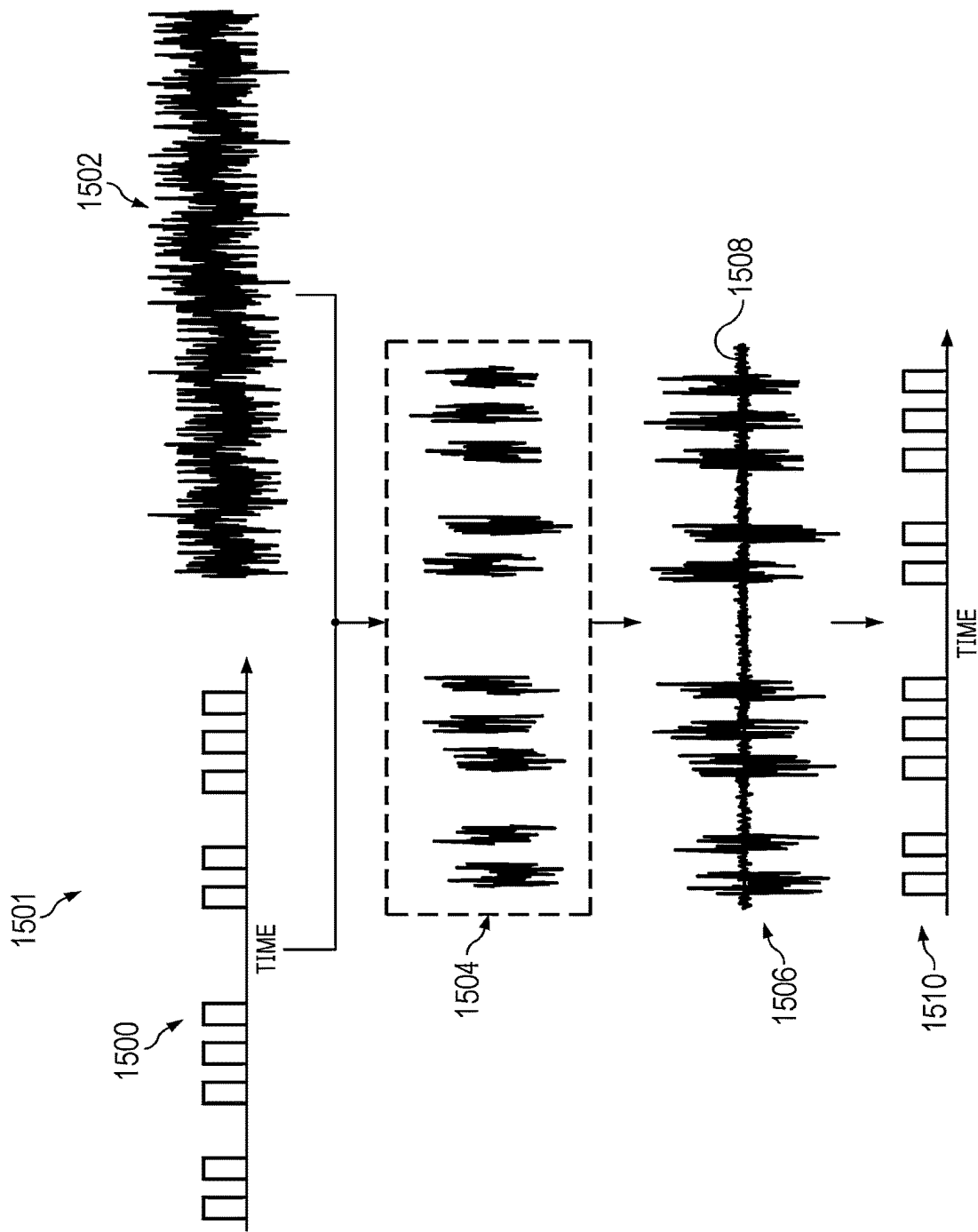


FIG. 15A

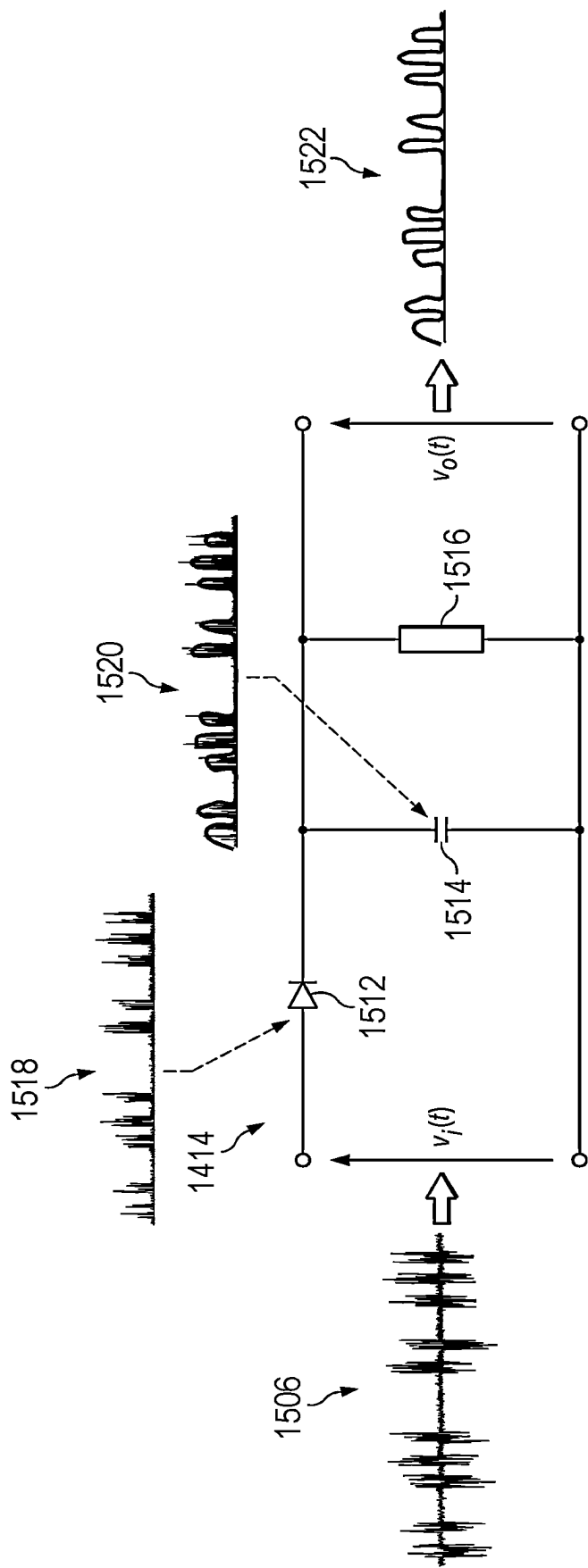


FIG. 15B

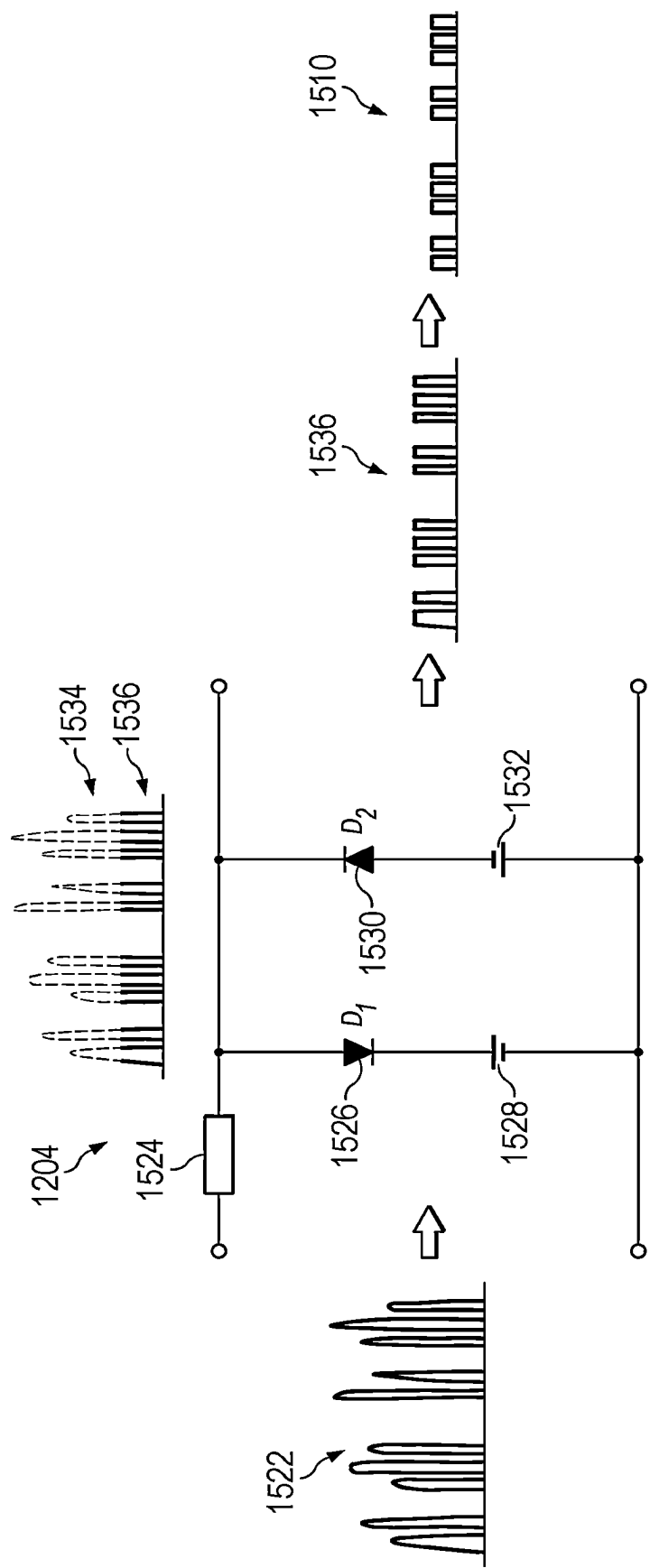


FIG. 15C

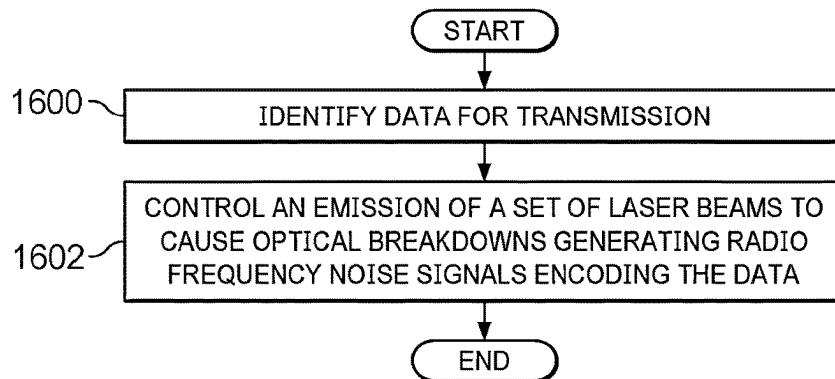


FIG. 16

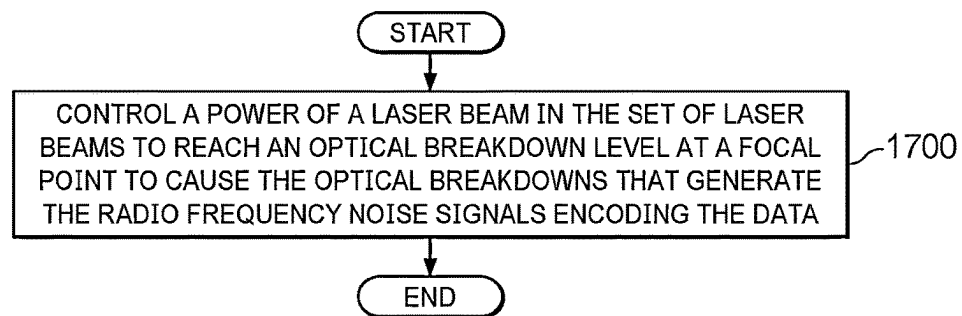


FIG. 17

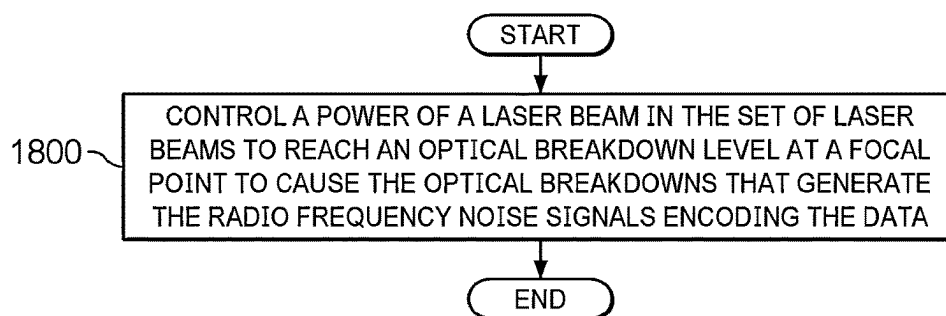


FIG. 18

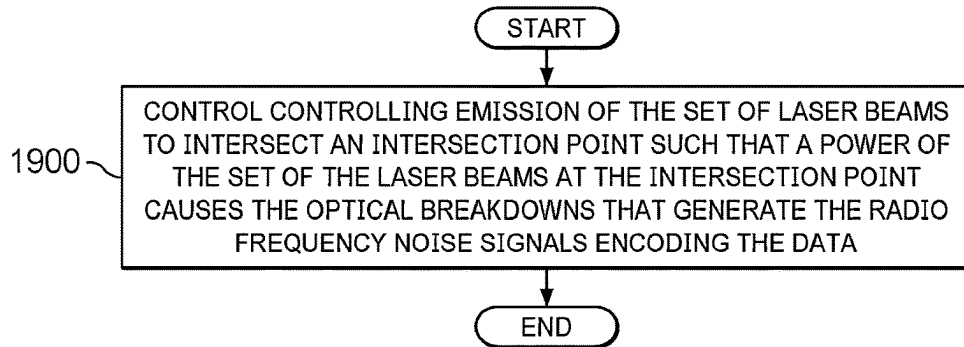


FIG. 19

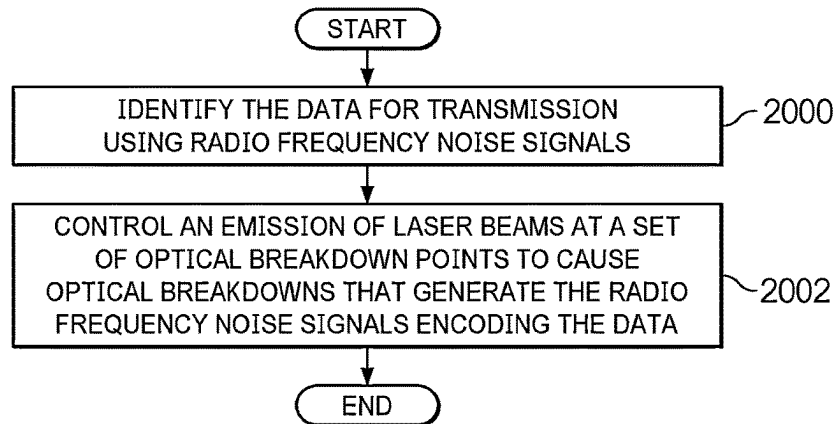


FIG. 20

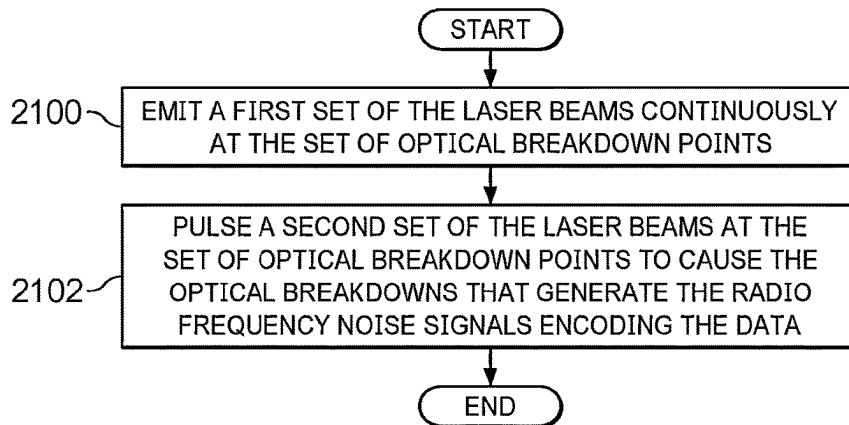


FIG. 21

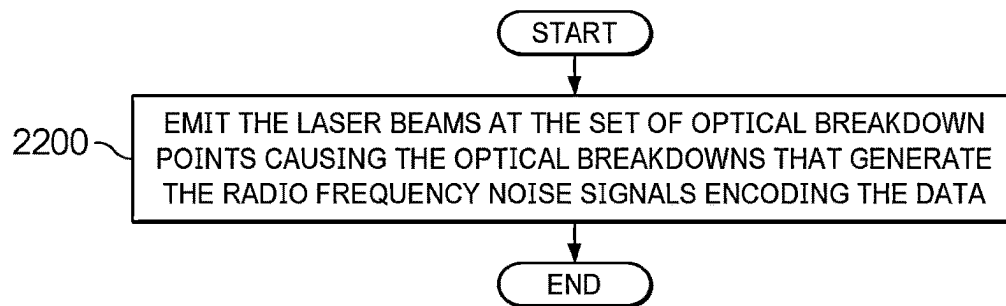


FIG. 22

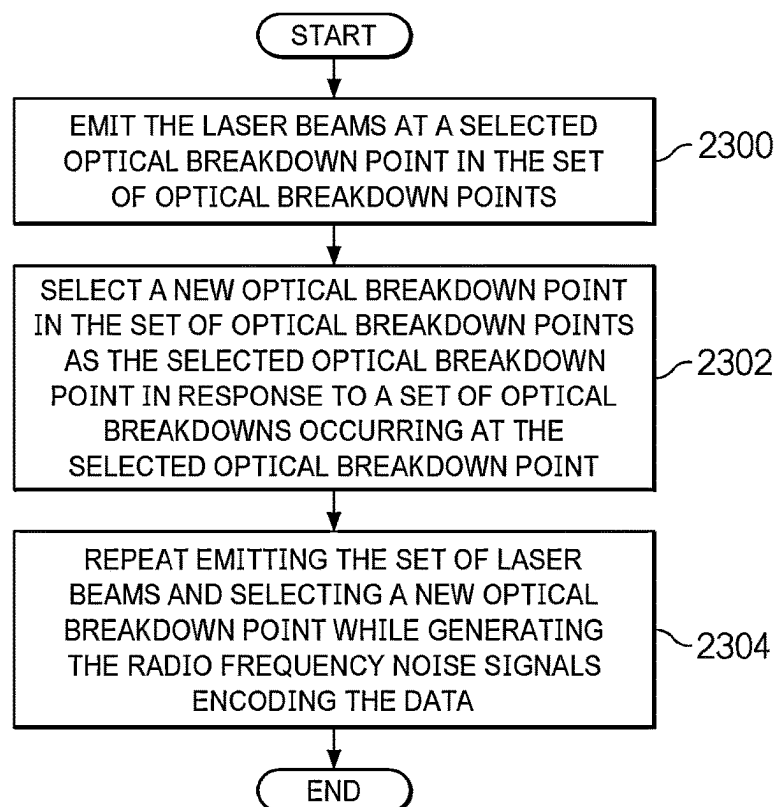


FIG. 23

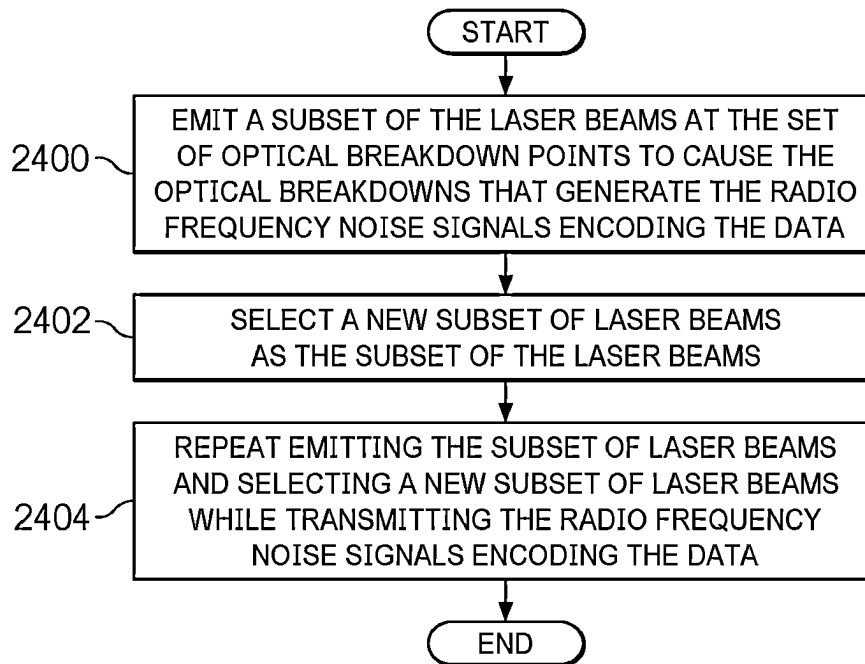


FIG. 24

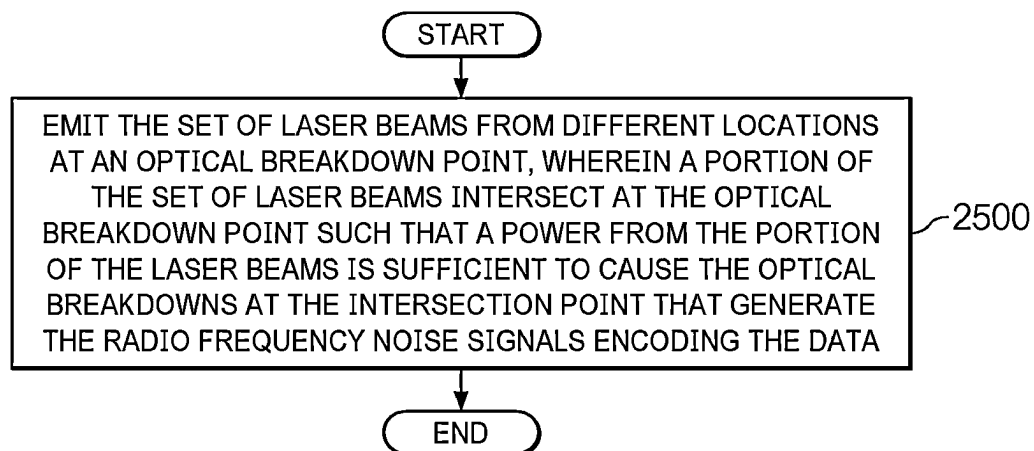


FIG. 25

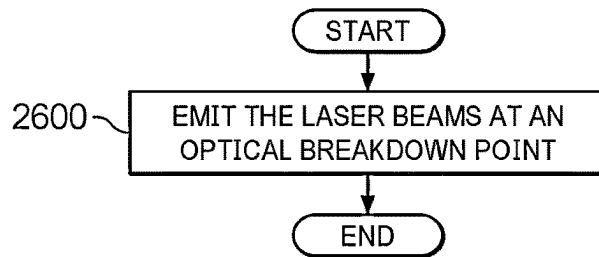


FIG. 26

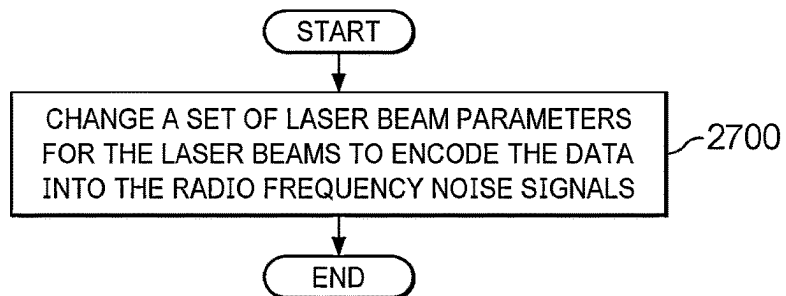


FIG. 27

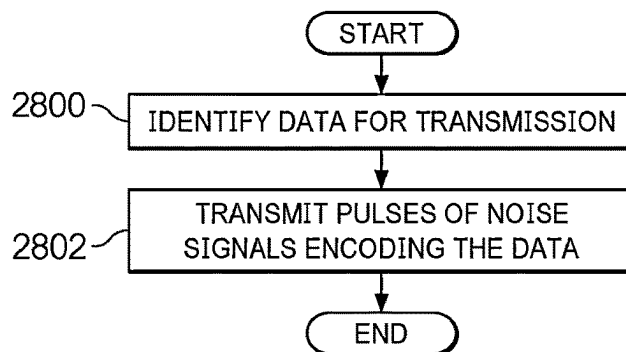


FIG. 28

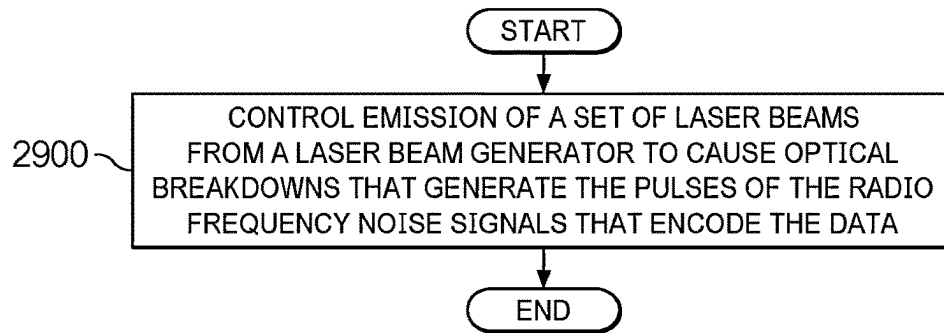


FIG. 29

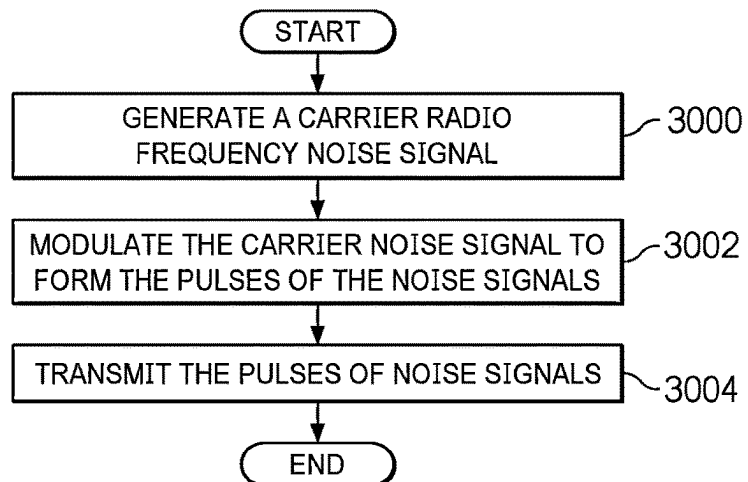


FIG. 30

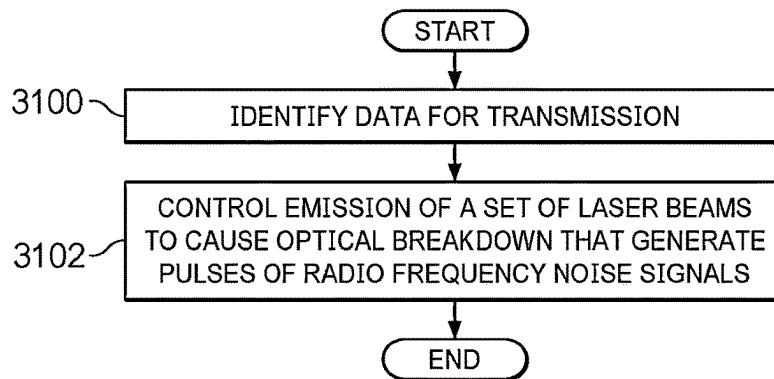


FIG. 31

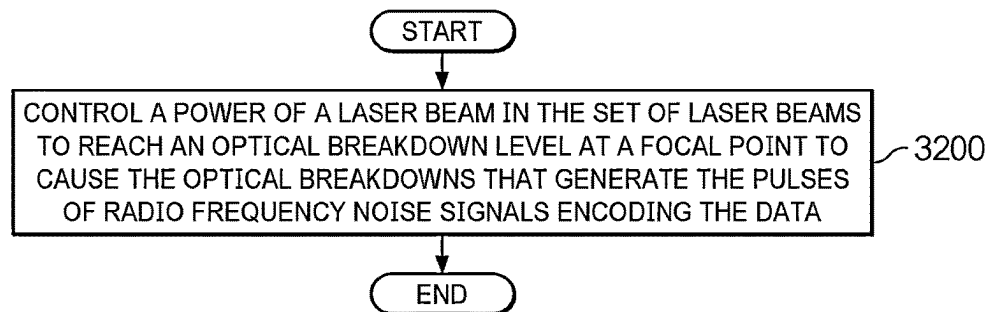


FIG. 32

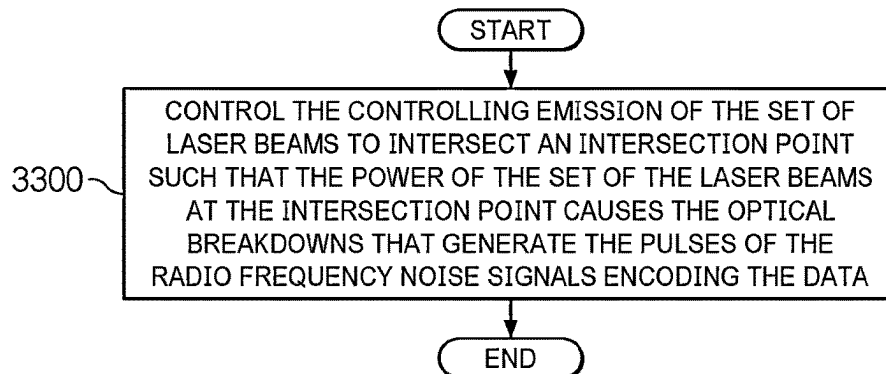


FIG. 33

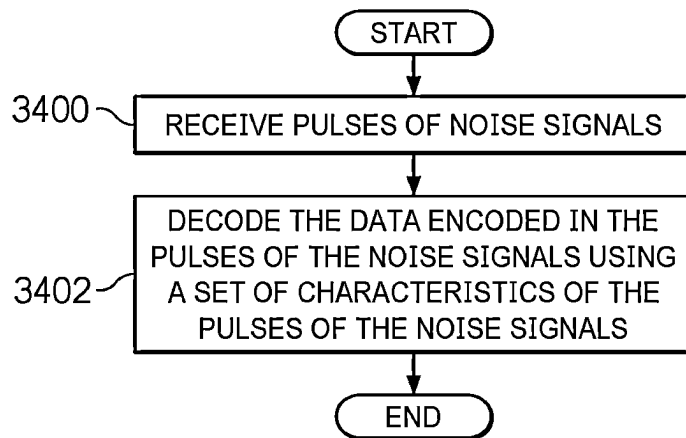


FIG. 34

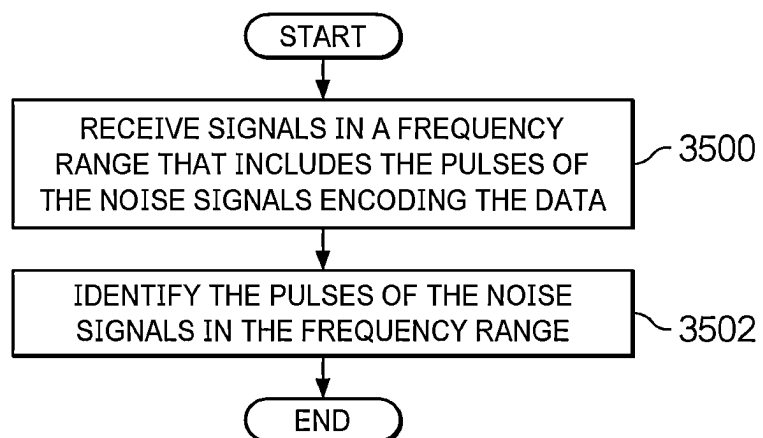


FIG. 35

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RADIO FREQUENCY COMMUNICATIONS USING LASER OPTICAL BREAKDOWNS

CROSS-REFERENCE TO RELATED APPLICATION

This application is related to the following U.S. Patent Application entitled “Pulse Noise Modulation to Encode Data”, Ser. No. 18/067,547, filed Dec. 16, 2022; filed even date hereof, assigned to the same assignee, and incorporated herein by reference in its entirety.

BACKGROUND INFORMATION

1. Field

The present disclosure relates generally to communications and in particular, to methods, apparatuses, systems, and computer program products for communicating data using radio frequency, optical, and/or other signals within the electromagnetic spectrum without physical antenna structures.

2. Background

Wireless communications using radio frequency signals, optical, and/or other signals within the electromagnetic spectrum are common and widespread. Radio frequency signals are commonly used in computer networks, for example, in the form of Wi-Fi signals that provide communications links between various computing devices.

Radio frequency signals are also used for communications between various clients such as ships, aircraft, land vehicles, buildings, and other locations. These communications can include data such as position information, voice messages, voice communications, and other types of data. For example, other types of data can include digital and analog signaling.

Communications using radio frequency transmissions are facilitated using physical antennas. The transmission or reception of radio frequency signals occurs between antennas. The use of physical antennas can be less convenient or reliable than desired.

In addition, radio frequency communications can be implemented using a carrier signal or carrier wave modulated by a modulation signal, message signal and/or information signal that modulates the carrier wave. Carrier signals use periodic waves, repeating waveforms, and/or other predictable waveforms such as sinusoidal, square-waves, sawtooth, or other repeatable carriers which are then modulated in various ways by the message signal, modulation signal, and/or information signal.

Therefore, it would be desirable to have methods, systems, and apparatuses that take into account at least some of the issues discussed above, as well as other possible issues. For example, it would be desirable to have methods and apparatus that overcome a technical problem with radio frequency communications using physical antennas. It would also be desirable to have systems, methods, and apparatuses that overcome the limitations of periodic and/or predictable carriers.

SUMMARY

An embodiment of the present disclosure provides a radio frequency communications system comprising a computer system and a communications manager in the computer

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system. The communications manager is configured to identify data for transmission using radio frequency noise signals. The communications manager is configured to control an emission of a set of laser beams at a set of optical breakdown points to cause optical breakdowns that generate radio frequency noise signals encoding the data.

Another embodiment of the present disclosure provides a radio frequency communications system comprising a laser generation system, a computer system, and a communications manager. The laser generation system is configured to emit a set of laser beams. The communications manager is configured to identify data for transmission. The communications manager is configured to control an emission of the set of laser beams using the laser generation system to cause optical breakdowns generating radio frequency noise signals encoding the data.

Still another embodiment of the present disclosure provides a method for transmitting data. The data is identified for transmission. An emission of a set of laser beams is controlled to cause optical breakdowns generating radio frequency noise signals encoding the data.

Yet another embodiment of the present disclosure provides a method for transmitting data. The data is identified for transmission using radio frequency noise signals. Emission of laser beams at a set of optical breakdown points is controlled to cause optical breakdowns that generate radio frequency noise signals encoding the data.

BRIEF DESCRIPTION OF THE DRAWINGS

The novel features believed characteristic of the illustrative embodiments are set forth in the appended claims. The illustrative embodiments, however, as well as a preferred mode of use, further objectives and features thereof, will best be understood by reference to the following detailed description of an illustrative embodiment of the present disclosure when read in conjunction with the accompanying drawings, wherein:

FIG. 1A is a pictorial representation of platforms that can transmit radio frequency signals using non-physical antennas in which illustrative embodiments may be implemented;

FIG. 1B is a pictorial representation of platforms that can transmit radio frequency signals using non-physical antennas from space in which illustrative embodiments may be implemented;

FIG. 2 is an illustration of a block diagram of a radio frequency communications environment in accordance with an illustrative embodiment;

FIG. 3 is an illustration of radio frequency noise generation using a laser beam in accordance with an illustrative embodiment;

FIG. 4 is an illustration of radio frequency noise generation using a plurality of laser beams in accordance with an illustrative embodiment;

FIG. 5 is an illustration of radio frequency noise generation using a plurality of laser beams in accordance with an illustrative embodiment;

FIG. 6 is an illustration of a diagram for controlling radio frequency noise generation in accordance with an illustrative embodiment;

FIG. 7 is an illustration of a diagram for controlling radio frequency noise generation in accordance with an illustrative embodiment;

FIG. 8 is an illustration of data transmission using pulse code noise modulation or pulse noise modulation in accordance with an illustrative embodiment;

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FIG. 9 is an illustration of a block diagram of a communication system in accordance with an illustrative embodiment;

FIG. 10 is an illustration of a transmitter in accordance with an illustrative embodiment;

FIG. 11 is an illustration of a block diagram of a radio frequency transmitter in accordance with an illustrative embodiment;

FIG. 12 is an illustration of a block diagram of a receiver in accordance with an illustrative embodiment;

FIG. 13 is an illustration of a communications system for transmitting and receiving electromagnetic noise signals in accordance with an illustrative embodiment;

FIG. 14 is an illustration of a block diagram of a communications system for transmitting and receiving electromagnetic noise signals in accordance with an illustrative embodiment;

FIG. 15A is an illustration of a data flow of signals transmitting data using modulated noise signals in accordance with an illustrative embodiment;

FIG. 15B is an illustration of an envelope follower circuit using a diode detector with a low pass filter in accordance with an illustrative embodiment;

FIG. 15C is an illustration of a clipper circuit in accordance with an illustrative embodiment;

FIG. 16 is an illustration of a flowchart of a process for transmitting data in accordance with an illustrative embodiment;

FIG. 17 is an illustration of a flowchart for controlling the emission of laser beams in points in accordance with an illustrative embodiment;

FIG. 18 is an illustration of a flowchart for controlling the emission of laser beams in points in accordance with an illustrative embodiment;

FIG. 19 is an illustration of a flowchart for controlling the emission of laser beams in points in accordance with an illustrative embodiment;

FIG. 20 is an illustration of a flowchart of a process for transmitting data in accordance with an illustrative embodiment;

FIG. 21 is an illustration of a flowchart for controlling the emission of laser beams in points in accordance with an illustrative embodiment;

FIG. 22 is an illustration of a flowchart for controlling the emission of the laser beams in points in accordance with an illustrative embodiment;

FIG. 23 is an illustration of a flowchart for controlling the emission of laser beams in points in accordance with an illustrative embodiment;

FIG. 24 is an illustration of a flowchart for controlling the emission of laser beams in points in accordance with an illustrative embodiment;

FIG. 25 is an illustration of a flowchart for controlling the emission of laser beams in points in accordance with an illustrative embodiment;

FIG. 26 is an illustration of a flowchart for controlling the emission of laser beams in points in accordance with an illustrative embodiment;

FIG. 27 is an illustration a flowchart for controlling laser beams in accordance with an illustrative embodiment;

FIG. 28 is an illustration of a flowchart for communicating data in accordance with an illustrative embodiment;

FIG. 29 is an illustration of a flowchart for transmitting pulses of noise signals in accordance with an illustrative embodiment;

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FIG. 30 is another illustration of a flowchart for transmitting pulses of noise signals in accordance with an illustrative embodiment;

FIG. 31 is an illustration of a flowchart for communicating data in accordance with an illustrative embodiment;

FIG. 32 is an illustration of a flowchart of a process for controlling the emission of a set of laser beams in accordance with an illustrative embodiment;

FIG. 33 is an illustration of a flowchart of a process for controlling the emission of a set of laser beams in accordance with an illustrative embodiment;

FIG. 34 is an illustration of a flowchart for communicating data in accordance with an illustrative embodiment; and

FIG. 35 is an illustration of a flowchart for decoding data in accordance with an illustrative embodiment.

DETAILED DESCRIPTION

The illustrative embodiments recognize and take into account one or more different considerations as described herein. For example, currently used physical antennas for transmitting radio frequency signals are subject to damage or destruction from various causes. For example, adverse weather conditions such as a hurricane or tornado can damage or destroy antennas such as transmission towers for land based communications. As another example, these physical antennas are also subject to damage or destruction from kinetic attacks.

Thus, the illustrative embodiments provide a method, apparatus, system, and computer program product for transmitting of radio frequency signals without hardware such as transmission towers and physical antenna structures. In one or more illustrative examples we provide a non-physical radio frequency antenna that is impervious to adverse environmental conditions and kinetic attack. Further, these non-physical radio frequency antennas can be more difficult to detect.

These transmitters can be positioned away from airplanes, transport, installations, buildings, or other locations that are subject to attack or undesired environmental conditions.

In the illustrative examples, radio frequency transmissions transmitted using laser beams that induce or cause optical breakdowns to generate the radio frequency transmissions. In this illustrative example, the optical breakdowns create plasma that generates the radio frequency signals. These optical breakdown points are the points of origination for transmitting the radio frequency signals.

With reference now to the figures and, in particular, with reference to FIG. 1A, a pictorial representation of platforms that can transmit radio frequency signals using non-physical antennas is depicted in which illustrative embodiments may be implemented. As depicted, radio frequency signals can be transmitted from various platforms as depicted in this figure.

As depicted, ground station 102 can transmit radio frequency signals 104 without using a physical antenna. In a similar fashion, ground station 106 can also transmit radio frequency signals 108 without using a physical antenna.

In this example, laser beams are used by these ground stations to transmit the radio frequency signals. For example, ground station 102 emits laser beam 110 in a manner that causes optical breakdown 112 at optical breakdown point 114. Radio frequency signals 104 are generated at and transmitted from optical breakdown point 114.

In this example, ground station 106 emits laser beam 116 and laser beam 118 at optical breakdown point 120 to cause optical breakdown 122. In this example, two laser beams are

used to cause optical breakdown **122** that results in transmission of radio frequency signals **108**.

This type of transmission can be used from other platforms such as train **130**. In this example, train **130** emits laser beam **132** and laser beam **134** from different locations on train **130** at optical breakdown point **136**. The intersection of these two laser beams at optical breakdown point **136** causes optical breakdown **138**. As a result, radio frequency signals **142** are transmitted in response to optical breakdown **138** at optical breakdown point **136**.

As another example, airplane **140** transmits radio frequency signals **142** using laser beam **144**. As depicted, laser beam **144** is emitted from airplane **140** at optical breakdown point **146**. Optical breakdown **148** occurs at optical breakdown point **146** which results in the transmission of radio frequency signals **142**.

Turning now to FIG. 1B, a pictorial representation of platforms that can transmit radio frequency signals using non-physical antennas from space in which illustrative embodiments may be implemented. As another example, in FIG. 1B, satellite **160** emits laser **164** from space into the atmosphere **180** above earth **190** while satellite **162** emits laser **166** from space into the atmosphere **180** such that laser **164** and laser **166** intersect at optical breakdown point **168** causing optical breakdown **170** which results in radio frequency signals **172** originating and emanating from optical breakdown point **168**.

As depicted, these radio frequency signals are generated without using physical antennas to transmit signals. Further, these radio frequency signals are transmitted at locations away from the platforms. As a result, identifying the platforms generating these radio frequency signals can be more difficult because antennas for transmitting the radio frequency signals are not visible. Further, tracking the location of where the radio frequency signals are generated does not provide identification of the platform or the platform location, nor the location of the communications system, computer system, communications manager, or the laser origin points in these examples.

The locations of these optical breakdowns are considered radio frequency source emitters that can be in remote locations from the platforms emitting the laser beams. As a result, identifying the locations of the platforms becomes more difficult with the absence of physical antennas.

Illustration of the different platforms in radio frequency communications environment **100** are only provided as examples of platforms that can implement this type of radio frequency signal transmission. In other illustrative examples, other platforms in addition to or in place of these platforms can be used. For example, this type of radio frequency generation can be implemented in a surface ship, a tank, or some other suitable type of platform in other illustrative examples.

With reference now to FIG. 2, an illustration of a block diagram of a radio frequency communications environment is depicted in accordance with an illustrative embodiment. In this illustrative example, radio frequency communications system **200** in communications environment **202** can communicate data **203** by using radio frequency signals **204** in the form of radio frequency noise signals **206**.

Data **203** can take a number of different forms. For example, data **203** can be a document, a spreadsheet, sensor data, an image, a video, and email message, a text message, a webpage, a table, a data structure, serial data, commands, or other types of data that is to be transmitted or communicated. Data can also be analog or digital data. Analog and digital data can include, for example, music and audio.

In one illustrative example, a noise signal is a signal with irregular fluctuations that are or appear to be at least one of random, non-predictable, or non-deterministic.

Further, the phrase “at least one of,” when used with a list of items, means different combinations of one or more of the listed items can be used, and only one of each item in the list may be needed. In other words, “at least one of” means any combination of items and number of items may be used from the list, but not all of the items in the list are required. The item can be a particular object, a thing, or a category.

For example, without limitation, “at least one of item A, item B, or item C” may include item A, item A and item B, or item B. This example also may include item A, item B, and item C or item B and item C. Of course, any combinations of these items can be present. In some illustrative examples, “at least one of” can be, for example, without limitation, two of item A; one of item B; and ten of item C; four of item B and seven of item C; or other suitable combinations.

A noise signal can be a signal that is statistically random. For example, a noise signal in these examples can be a signal that meets one or more standard tests for statistical randomness. A pseudorandom noise signal that seems to lack any definite pattern, although consisting of a deterministic sequence of pulses that repeats itself after its period is an example of a signal that is statistically random and considered a noise signal that can be used to encode data.

In this example, radio frequency noise signals **206** are electromagnetic noise signals that can have a frequency from around 20 kHz to above the Terahertz range. Radio frequency noise signals can include signals with frequencies such as extremely low frequency (ELF), high frequency (HF), and other types of frequencies. These noise signals can also include microwave noise signals and Terahertz noise signals. Electromagnetic noise signals can also be optical noise in the visible range, infrared, ultraviolet X-rays and other types of noise signals that can be used as modulated noise. For example, lasers used at optical breakdown also may transmit as various ranges of noisy light in addition to noisy broadband radio frequencies. Modulating this noisy light with different techniques such as pulse noise modulation is included in this disclosure.

In this illustrative example, radio frequency communications system **200** is associated with platform **208**. Platform **208** is an object that can transmit radio frequency noise signals **206** using radio frequency communications system **200**.

Platform **208** can take a number of different forms. For example, platform **208** can be one of a mobile platform, a stationary platform, a land-based structure, an aquatic-based structure, and a space-based structure. More specifically, the platform can be a surface ship, a tank, a personnel carrier, a train, an airplane, a commercial airplane, a spacecraft, a space station, a satellite, a submarine, an automobile, a ground station, a power plant, a bridge, a dam, a house, a manufacturing facility, a building, and other suitable platforms.

In this illustrative example, radio frequency communications system **200** comprises computer system **210** and communications manager **212**. In this example, communications manager **212** is located in computer system **210**.

As used herein, “a number of” when used with reference to items, means one or more items. For example, “a number of different forms” is one or more different forms.

Communications manager **212** can be implemented in software, hardware, firmware or a combination thereof. When software is used, the operations performed by com-

munications manager **212** can be implemented in program instructions configured to run on hardware, such as a processor unit. When firmware is used, the operations performed by communications manager **212** can be implemented in program instructions and data and stored in persistent memory to run on a processor unit.

When hardware is employed, the hardware may include circuits that operate to perform the operations in communications manager **212**. The circuits used to implement communications manager **212** can take other forms in addition to or in place of a processor unit.

In the illustrative examples, the hardware used to implement communications manager **212** can take a form selected from at least one of a circuit system, an integrated circuit, an application specific integrated circuit (ASIC), a programmable logic device, or some other suitable type of hardware configured to perform a number of operations. With a programmable logic device, the device can be configured to perform a number of operations. The device can be reconfigured at a later time or can be permanently configured to perform the number of operations. Programmable logic devices include, for example, a programmable logic array, a programmable array logic, a field programmable logic array, a field programmable gate array, and other suitable hardware devices. Additionally, the processes can be implemented in organic components integrated with inorganic components and can be comprised entirely of organic components excluding a human being. For example, the processes can be implemented as circuits in organic semiconductors.

Computer system **210** is a physical hardware system and includes one or more data processing systems. In this illustrative example, the data processing systems are hardware machines that can be configured to perform a sequence of operations. These operations can be performed in response to receiving an input in generating and output based on performing the operations. This output can be data in the form of values, commands, or other types of data. When more than one data processing system is present in computer system **210**, those data processing systems are in communication with each other using a communications medium. The communications medium may be a network. The data processing systems may be selected from at least one of a computer, a server computer, a tablet, or some other suitable data processing system.

As depicted, computer system **210** includes a number of processor units **214** that are capable of executing program instructions **216** implementing processes in the illustrative examples. In other words, program instructions **216** are computer readable program instructions.

As used herein, a processor unit in the number of processor units **214** is a hardware device and is comprised of hardware circuits such as those on an integrated circuit that respond and process instructions and program code that operate a computer. When the number of processor units **214** executes program instructions **216** for a process, the number of processor units **214** can be one or more processor units that are on the same computer or on different computers. In other words, the process can be distributed between processor units **214** on the same or different computers in a computer system **210**.

Further, the number of processor units **214** can be of the same type or different type of processor units. For example, a number of processor units **214** can be selected from at least one of a single core processor, a dual-core processor, a multi-processor core, a general-purpose central processing unit (CPU), a graphics processing unit (GPU), a digital signal processor (DSP), or some other type of processor unit.

As depicted, radio frequency communications system **200** can also include laser generation system **218**. In other examples, laser generation system **218** can be considered a separate component controlled by radio frequency communications system **200**.

In this example, laser generation system **218** is a hardware system that can emit a set of laser beams **220**. The operation of laser generation system **218** can be controlled by communications manager **212**.

In this example, the set of laser beams **220** can be emitted from different locations **221**. For example, laser generation system **218** can be comprised of laser units that are positioned in different locations. Each location can have one or more laser units for laser generation system **218** in this illustrative example.

Communications manager **212** can identify data **203** for transmission using radio frequency noise signals **206**. Communications manager **212** controls an emission of a set of laser beams **220**. In this example, communications manager **212** directs or steers the set of laser beams **220** at a set of optical breakdown points **222**. In this example, the set of optical breakdown points **222** can be selected from at least one of intersection point **223** or focal point **225**.

As used herein, “a set of” when used with reference to items, means one or more items. For example, “a set of optical breakdown points **222**” is one or more of optical breakdown points **222**. In another example, a “set of laser beams” means one or more laser beams.

In this example, intersection point **223** can be a location where two or more laser beams intersect. This location can be where an optical breakdown occurs from the intersection of two or more laser beams when the power **227** of two or more intersecting laser beams is sufficient to cause an optical breakdown. Focal point **225** can be a location where the laser beam is focused to cause an optical breakdown to occur at that location.

This emission of the set of laser beams **220** is controlled by communications manager **212** to cause optical breakdowns **224** at the set of optical breakdown points **222** that generate radio frequency noise signals **206** encoding data **203**. In this illustrative example, plasma **226** occurs at optical breakdown points **222** in response to optical breakdowns **224** by the set of laser beams **220**. This plasma generated by optical breakdowns **224** causes radio frequency noise signals **206** to be transmitted at the set of optical breakdown points **222**.

In this example, power **227** of laser beam **228** in the set of laser beams **220** at optical breakdown point **230** in the set of optical breakdown points **222** can be controlled using different mechanisms. For example, power **227** can be controlled by at least one of a shutter, a lens, a deformable lens, a microelectromechanical systems mirror, an attenuator, a controlling optics, an optical filter, an amplitude modulator in a laser beam generator, or other device.

In this illustrative example, communications manager **212** can control the emission of the set of laser beams **220** by laser generation system **218** in a number of different ways. For example, communications manager **212** can control laser generation system **218** to emit a first number of the set of laser beams **220** continuously at the set of optical breakdown points **222**. Communications manager **212** can control laser generation system **218** to pulse a second number of the set of laser beams **220** at the set of optical breakdown points **222** to cause optical breakdowns **224** that generate radio frequency noise signals **206** encoding data **203**. The laser beam can be pulsed by turning the laser beam on and off. In another example, a laser beam can be pulsed by varying the

power of the laser beam. In other words, the power can be pulsed by increasing and decreasing the power of the laser beam.

In another illustrative example, communications manager **212** can control laser generation system **218** to emit the set of laser beams **220** at the set of optical breakdown points **222** causing optical breakdowns **224** that generate radio frequency noise signals **206** encoding data **203**.

In this example, the emission of the set of laser beams **220** can be performed in a number of different ways. The set of laser beams can be emitted as at least one of as pulsed or continuous. For example, one laser beam can be continuous while another laser beam is pulsed. Further, the laser beams can be originated from different directions at the set of optical breakdown points **222**.

The direction at which a laser beam is emitted can move or sweep back such that an optical breakdown point is included during the movement of the laser beam. In other words, during the sweeping of the laser beam the laser beam can intersect with another laser beam. The intersection of this laser beam with another laser beam emitted the optical breakdown point can cause the optical breakdown at that optical breakdown point.

In another illustrative example, communications manager **212** can control laser generation system **218** to emit the set of laser beams **220** at selected optical breakdown point **232** in the set of optical breakdown points **222**. Communications manager **212** can select new optical breakdown point **234** in the set of optical breakdown points as the selected optical breakdown point. Communications manager **212** can repeat emitting the set of laser beams **220** and selecting the new optical breakdown point while generating radio frequency noise signals **206** encoding data **203**.

In yet another illustrative example, communications manager **212** can control laser generation system **218** to emit the set of laser beams **220** from different locations **221** at optical breakdown point **230**. In this example, a portion of the set of laser beams **220** intersect at optical breakdown point **230** such that power **227** from the portion of the set of laser beams **220** is sufficient to cause optical breakdowns **224** at intersection point **223** that generate radio frequency noise signals **206** encoding data **203**.

As another example, communications manager **212** can control laser generation system **218** to emit the set of laser beams **220** at optical breakdown point **230**. In this example, optical breakdowns **224** occur in response to all of the set of laser beams **220** intersecting at optical breakdown point **230**.

In controlling the emission of the set of laser beams **220**, communication manager **212** can change a set of laser beam parameters **240** for the set of laser beams **220** to encode data **203** into radio frequency noise signals **206**. In this example, changing the set of laser beam parameters **240** changes a set of radio frequency characteristics **242** for radio frequency noise signals **206**. The set radio frequency characteristics **242** for radio frequency noise signals **206** can be selected from at least one of a timing, an optical breakdown point, an amplitude of the radio frequency noise signals, a frequency band, a relative phase, or other characteristics for radio frequency noise signals **206**.

In yet another illustrative example, communications manager **212** can control laser generation system **218** to emit a subset of the set of laser beams **220** at the set of optical breakdown points **222** to cause the optical breakdowns **224** that generates radio frequency noise signals **206** encoding data **203**. Communications manager **212** can select a new subset of the set of laser beams **220** as the subset of laser beams **220**.

Communications manager **212** can repeat emitting of the subset of the set of laser beams **220** and selecting the new subset of the set of laser beams **220** while transmitting radio frequency noise signals **206** encoding the data **203**.

Thus, one or more illustrative examples enable transmitting radio frequency noise signals using radio frequency source emitters that do not require physical structures. As a result, one or more illustrative examples can overcome an issue with the vulnerability present in using physical source emitters such as antennas. In the illustrative examples, the optical breakdown points for the optical breakdowns are radio frequency source emitters.

Further, these radio frequency source emitters can be moved almost instantaneously to different locations by repositioning the laser beams such that the laser beams point at different optical breakdown points. Attacks at these radio frequency source locations are attacks at the optical breakdown points where the plasma is generated. As a result, kinetic attacks against these locations are useless because the laser modulation sources are remote from the locations of these radio frequency source emitters.

The illustration of communications environment **202** to in FIG. **2** is not meant to imply physical or architectural limitations to the manner in which an illustrative embodiment may be implemented. Other components in addition to or in place of the ones illustrated may be used. Some components may be unnecessary. Also, the blocks are presented to illustrate some functional components. One or more of these blocks may be combined, divided, or combined and divided into different blocks when implemented in an illustrative embodiment.

For example, although communications manager **212** is shown as being implemented using program instructions **216** run on a number of processor units **214** in computer system **210**, communications manager **212** can be implemented in other hardware instead of or in addition to the number of processor units **214**. For example, computer system **210** may use other hardware in addition to or in place of the number of processor units **214**.

For example, other types of hardware circuits capable of performing the operations for communications manager **212** can be used. This other hardware can be at least one of a circuit system, an integrated circuit, an application specific integrated circuit (ASIC), a programmable logic device, or some other suitable type of hardware configured to perform a number of operations.

Turning next to FIG. **3**, an illustration of radio frequency noise generation using a laser beam is depicted in accordance with an illustrative embodiment. This depicted example in FIG. **3** illustrates how a single laser beam can be used to generate radio frequency signals.

In this illustrative example, laser generation system **300** is an example of laser generation system **218** in FIG. **2**. Laser generation system **300** emits laser beam **302**.

As depicted, laser generation system **300** comprises a number of different components. In this example, laser generation system **300** includes oscillator **303** and optical system **304**.

Oscillator **303** generates coherent light for emitting laser beam **302**. In this example, optical system **304** can focus laser beam **302**. Optical system **304** includes at least one of a lens, a mirror, or other optical element that can change the focus of laser beam **302**.

In this example, the focus of laser beam **302** is controlled such that the power at focal point **306** is an optical breakdown point **308** where optical breakdown **310** occurs. As depicted in this example, optical breakdown **310** results in

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the generation of plasma 312. Plasma 312 resulting from optical breakdown 310 causes the generation of radio frequency noise signal 314. Thus, this example illustrates how a single laser beam can be used to generate radio frequency signals.

Turning next to FIG. 4, an illustration of radio frequency noise generation using a plurality of laser beams is depicted in accordance with an illustrative embodiment. In this illustrative example, laser generation system 420 comprises laser unit 400 and laser unit 402. Laser generation system 420 is an example of an implementation for laser generation system 218 in FIG. 2.

In this illustrative example, laser unit 400 generates first laser beam 404. Laser unit 402 generates second laser beam 406.

In this example, first laser beam 404 and second laser beam 406 are emitted in directions from these laser beam units to intersect at optical breakdown point 408. These two laser beams are emitted along different paths that intersect at optical breakdown point 408. This optical breakdown point where the two laser beams intersect each other is intersection point 410.

In this example, the intersection of first laser beam 404 and second laser beam 406 results in optical breakdown 412. This optical breakdown generates plasma 414. As depicted in this example, optical breakdown 412 results in radio frequency noise signals 416.

As depicted in the example, optical breakdown 412 occurs where first laser beam 404 and second laser beam 406 intersect at intersection point 410. In this example, the power for first laser beam 404 and second laser beam 406 individually is not sufficient to cause an optical breakdown.

The illustration of the two laser units for laser generation system 420 in FIG. 4 is provided as an example of one implementation for generating radio frequency noise signals. This illustration is not meant to limit the manner in which other illustrative examples can be implemented. In another example, one or more laser units in addition to laser unit 400 and laser unit 402 can be used to generate additional laser beams. The laser beams can also intersect at intersection point 410 to cause optical breakdown 412.

Turning next to FIG. 5, an illustration of radio frequency noise generation using a plurality of laser beams is depicted in accordance with an illustrative embodiment. In this illustrative example, laser generation system 520 comprises laser unit 500 and optical system 502. Laser generation system 520 is an example of an implementation for laser generation system 218 in FIG. 2.

In this example, laser unit 500 emits first laser beam 504 and second laser beam 506. In this example, laser unit 500 generates initial laser beam 501 that is split into two laser beams, first laser beam 504 and second laser beam 506 by optical system 502.

As depicted, optical system 502 comprises a number of different components. In this depicted example, optical system 502 comprises shutter 503, variable attenuator 505, beam splitter 507, mirror 509, and lens 511.

The components depicted are example components that can be used in optical system 502 and can change in other illustrative examples. For example, one or more of lens 511, variable attenuator 505, and shutter 503 may be omitted in other illustrative examples. In yet other illustrative examples, other components may be added such as a lens located before beam splitter 507.

As depicted, initial laser beam 501 is split into two laser beams by beam splitter 507. Mirror 509 can be used to direct second laser beam 506 in different directions. Further, mirror

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509 can be used to provide focus to increase the power of second laser beam 506 at a focal point such as optical breakdown point 508. Lens 511 also can be used to provide focus to increase the power of second laser beam 506 at optical breakdown point 508.

In this example, first laser beam 504 and second laser beam 506 are emitted in directions to intersect at optical breakdown point 508, which is intersection point 510 in this example. Optical breakdown 512 occurs at this intersection of first laser beam 504 and second laser beam 506, generating plasma 514 that results in the generation of radio frequency noise signals 516.

In this example, the power of first laser beam 504 and second laser beam 506 are sufficient to cause optical breakdown 512 at the intersection of the laser beams. Optical breakdowns do not occur in other locations where these laser beams do not intersect each other in this example.

With reference now to FIG. 6, an illustration of a diagram for controlling radio frequency noise generation is depicted in accordance with an illustrative embodiment. In this illustrative example, the operation of laser generation system 620 is controlled by controller 644. As depicted, laser generation system comprises laser unit 600, laser unit 602, first power source 640, second power source 642, and optical system 605. In this example, controller 644 can control the operation of laser generation system 620.

In this example, laser generation system 620 is an example of an implementation for laser generation system 218 in FIG. 2. Controller 644 is an example a component that can be implemented in communications manager 212 in FIG. 2.

In this illustrative example, controller 644 can control the emission of first laser beam 604 and second laser beam 606 from laser generation system 620. In this illustrative example, laser unit 600 generates first laser beam 604 using power supplied by first power source 640. Laser unit 602 generates second laser beam 606 using power supplied by second power source 642.

In this example, first laser beam 604 and second laser beam 606 are emitted in directions that have paths that intersect at optical breakdown point 608, which is intersection point 610. Optical breakdown 612 occurs at this intersection of first laser beam 604 and second laser beam 606, generating plasma 614 that results in the transmission of radio frequency noise signals 616.

In this example, controller 644 can control the emission of these laser beams such that at least one of first laser beam 604 or second laser beam 606 is pulsed. This pulsing can include at least one of turning a laser beam on and off for increasing and decreasing the power of the laser beam. This pulsing of one or both of first laser beam 604 and second laser beam 606 can be controlled to control the timing of radio frequency noise generation.

When pulsed, optical breakdown 612 occurs when both laser beams intersect at intersection point 610. When one laser beam is turned off, and intersection is not present between both laser beams and optical breakdown 612 does not occur. By controlling the timing of when first laser beam 604 and second laser beam 606 intersect at intersection point 610, controller 644 can control the generation of radio frequency noise signals in a manner that encodes data.

For example, data can be encoded in radio frequency noise signals based on the timing of when radio frequency noise signals are generated. As another example, the timing of the laser beams can be used to control the duration of radio frequency noise signals. This duration can also be used to encode data into the radio frequency noise signals.

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In this illustrative example, controller 644 can control whether a laser unit emits a continuous laser beam or a pulsed laser beam using components such as first power source 640 and second power source 642. These power sources can be turned on and off to turn the laser beams on and off. With this pulsing, optical breakdowns occur when both laser beams are on and intersect at intersection point 610.

In this example, the pulsing can also include increasing and decreasing the power in one or both of first laser beam 604 and second laser beam 606. In this example, decreasing the power of one or both laser beams can prevent the occurrence of an optical breakdown because of insufficient power being present when first laser beam 604 and second laser beam 606 intersect at intersection point 610. Optical breakdown 612 occurs when the power present from both laser beams intersecting at intersection point 610 is high enough for an optical breakdown.

As another example, the pulsing of the laser beams can also be controlled using optical elements in optical system 605. These optical elements can be controlled by controller 644 to pulse one or more of first laser beam 604 and second laser beam 606.

For example, variable attenuator 611 and shutter 613 can be operated to pulse first laser beam 604. For example, shutter 613 can be used to selectively emit first laser beam 604. Variable attenuator 611 can be used to change the power of first laser beam 604. In similar fashion, the emission of second laser beam 606 can also be pulsed using variable attenuator 615 and shutter 617.

Thus, the emission of first laser beam 604 and second laser beam 606 from laser generation system 620 can be controlled by controller 644 such that both laser beams are continuous, one laser beam is continuous while the other laser beam is pulsed, or both laser beams are pulsed. This control can be performed to achieve optical breakdowns to transmit radio frequency noise signals in a manner that encodes data into the radio frequency signals.

The illustration of laser generation system 620 is an example of one implementation and is not meant to limit the manner in which other illustrative examples can be implemented. For example, in other illustrative examples one or more laser units can be present in addition to laser unit 600 and laser unit 602.

With reference next to FIG. 7, an illustration of a diagram for controlling radio frequency noise generation is depicted in accordance with an illustrative embodiment. In this illustrative example, laser generation system 720 is controlled by controller 744. As depicted, laser generation system 720 comprises laser unit 700, power source 740, and optical system 702.

Laser generation system 720 is an example of an implementation for laser generation system 218 in FIG. 2. Controller 744 is an example of the components that can be implemented in communications manager 212 in FIG. 2.

In this illustrative example, controller 744 controls the emission of first laser beam 704 and second laser beam 706 from laser generation system 720. In this illustrative example, laser unit 700 generates first laser beam 704 and second laser beam 706 using power supplied by power source 740. In this example, laser unit 700 generates initial laser beam 701 that is split into two laser beams, first laser beam 704 and second laser beam 706 by optical system 702.

As depicted, optical system 702 comprises a number of different components. In this example, optical system 702 comprises shutter 703, variable attenuator 705, beam splitter 707, and mirror 709, mirror 711, mirror 713, and lens 715 as

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other components that can be located in optical system 702. The components depicted are example components that can be used in optical system 702 and these components can change in other illustrative examples. For example, one or more lens 715, variable attenuator 705, and shutter 703 may be omitted in other illustrative examples. In yet other illustrative examples, other components may be included such as a lens located before beam splitter 707.

As depicted, initial laser beam 701 is split into two laser beams by beam splitter 707. In this example, first laser beam 704 and second laser beam 706 are emitted in directions to intersect at optical breakdown point 708, which is intersection point 710 in this example. Optical breakdown 712 occurs at this intersection of first laser beam 704 and second laser beam 706. Optical breakdown 712 generates radio frequency signal 716 through plasma 714 occurring from optical breakdown 712.

In this example, the power of the first laser beam 704 and second laser beam 706 are sufficient to cause optical breakdown 712 at the intersection of the laser beams. Optical breakdowns do not occur in other locations where these laser beams do not intersect each other in this example.

In this example, controller 744 can control the emission of these laser beams such that at least one of first laser beam 704 or second laser beam 706 is pulsed. This pulsing can include at least one of turning a laser beam on and off for increasing and decreasing the power of the laser beam. This pulsing of one or both of first laser beam 704 and second laser beam 706 can be controlled to control the timing of radio frequency noise generation.

In this example, first laser beam 704 can be pulsed by controller 744 controlling the operation of at least one of variable attenuator 705 or shutter 703. Variable attenuator 705 can be used to change the power of first laser beam 704. Shutter 703 can turn laser beam on and off with respect to emissions of laser beams from laser generation system 720. In this example, both laser beams can be pulsed at the same time by controlling power source 740. In another illustrative example, components within laser unit 700 such as an amplitude modulator can be controlled to pulse the power of initial laser beam 701 resulting in a pulsing of both first laser beam 704 and second laser beam 706.

By controlling the timing of when first laser beam 704 and second laser beam 706 intersect at intersection point 710, controller 744 can control the generation of radio frequency noise signals in a manner that encodes data.

In yet another illustrative example, controller 744 can control the location of optical breakdown point 708 by moving one or both of first laser beam 704 and second laser beam 706. This movement of optical breakdown point 708 can be controlled using at least one of mirror 709 or mirror 713. By moving the location of optical breakdown point 708, the phase of radio frequency noise signal can be changed to encode data.

The illustration of example implementations for laser generation system 218 in FIG. 2 and in FIGS. 3-7 have been provided as an example of some illustrative examples and are not meant to limit the manner in which other laser generation systems can be implemented. For example, a laser generation system can include both a first laser unit and a second laser unit with an optical system. In yet another illustrative example, one or more laser units can be present that emit laser beams in addition to the ones depicted at different optical breakdown points. With this example, two or more optical breakdowns can occur from laser beams emitted from a laser generation system.

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In yet another illustrative example, different laser beams can be emitted at different times at the same optical breakdown point. As a result, optical breakdowns can be generated from different combinations of laser beams at the same optical breakdown point.

The illustrative embodiments also recognize and take into account that current techniques for transmitting data involves the use of carrier wave forms. For example, many techniques use only periodic, sinusoidal, or other repetitive or predictable carrier wave forms that are modulated to encode data. These types of waveforms can be detected in noise through various techniques including the denoiser technology which can detect sinusoidal carriers at 20 dB to 40 dB below a noise floor.

As a result, interception and decoding of signals can occur using current transmission techniques. Further, when the sinusoidal carriers can be detected, security issues can arise. For example, information can be inserted into transmissions, jamming attacks can occur, or other issues with using single sinusoidal, periodic, or other repetitive carriers to transmit data.

Thus, the illustrative embodiments provide a method, apparatus, and system for transmitting data. In the illustrative examples, this data can be transmitted using various modulation techniques that modulate noise signals. The use of noise signals is in contrast to the use of a sinusoidal, periodic, repetitive, or predictable carrier that can be detected.

Turning to FIG. 8, an illustration of data transmission using pulse code noise modulation (also called pulse noise modulation) is depicted in accordance with an illustrative embodiment. In this illustrative example, pulse code noise modulation or pulse noise modulation can be performed using a radio frequency communication system such as radio frequency communication system 200 in FIG. 2.

In this illustrative example, optical breakdowns are generated over time. These optical breakdowns result in the generation of plasma 800 that causes radio frequency noise signals 802 to be transmitted.

The timing of these optical breakdowns can be selected to encode data such that the generation of radio frequency noise signals 802 encode the data. In this example, pulses are present in radio frequency noise signals 802 with timing that corresponds to the timing of optical breakdowns that generated plasma 800. These pulses of radio frequency noise signals 802 are timed to encode data. This type of encoding of data can be referred to as pulse noise modulation. As depicted, radio frequency noise signals 802 can be received and decoded to obtain decoded data signal 804.

This illustration of using radio frequency noise signals generated by optical breakdowns to communicate data is presented as one example of how pulses of radio frequency noise signals can encode data. This illustration is not meant to limit the manner in which other illustrative examples can be implemented.

For example, the pulses of radio frequency noise signals can be generated using other techniques in addition to or in place of laser-induced optical breakdowns. A transmitter system can use a noise signal as a carrier signal and a modulator to modulate the carrier signal such that pulses of radio frequency noise are transmitted that encode the data.

In still other illustrative examples, other types of noise signals in addition to or in place of radio frequency electromagnetic noise signals can be used. For example, noise signals can be used for transmitting data encoded in pulses and can be selected from at least one of electromagnetic frequency noise signals, radio frequency noise signals,

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microwave frequency signals, audio frequency noise signals, ultrasonic frequency noise signals, ultra-low frequency noise signals, very low frequency noise signals, underwater frequency noise signals, optical frequency noise signals, or other types of noise signals. These different types of noise signals can be used for various applications including speech communication, music, or other types of information for data that that are encoded in the noise signals.

With reference next to FIG. 9, an illustration of a block diagram of a communication system is depicted in accordance with an illustrative embodiment. In this illustrative example, communications system 900 in communications environment 901 operates to transmit data 902 encoded in noise signals 904.

In one illustrative example, a noise signal is a signal with irregular fluctuations that are or appear to be random, non-predictable, or non-deterministic. A noise signal can be a signal that is statistically random. For example, a noise signal in these examples can be a signal that meets one or more standard tests for statistical randomness. A pseudorandom noise signal that seems to lack any definite pattern, although consisting of a deterministic sequence of pulses that repeats itself after its period is an example of a signal that is statistically random and considered a noise signal that can be used to encode data. In this example, the noise in noise signals 904 can be selected from at least one of nondeterministic noise, pseudo random noise, or some other suitable type of noise signal.

In the illustrative example, signals can have characteristics selected from at least one of amplitude, frequency, bandwidth, timing, phase, or other characteristics. In this illustrative example, noise signals 904 can be noise signals in which at least one of these characteristics are not controlled to encode the data. In other words, at least one or more of these characteristics meet one or more standard tests for statistical randomness in noise signals 904.

In these examples, noise signals 904 do not include carrier waves that are periodic. These types of signals can be, for example, sinusoidal, sawtooth, square, or other types of signals. Noise signals 904 also do not include periodic or sinusoid-based carrier signals that employ spread spectrum, frequency-hopping signals, and radar "chirps" that are based on periodic signals such as sinusoids or sawtooths. These and other types of signals that do not meet one or more standard tests for statistical randomness are not considered noise signals 904 in this example. However, "spread noise spectrum", frequency-hopping noise signals, and noise-based radar bursts that use noise as the basis of their carrier signals are considered noise signals 904 in this example.

As depicted, communications system 900 comprises computer system 910 and communications manager 912 located in computer system 910.

Communications manager 912 can be implemented in software, hardware, firmware or a combination thereof. When software is used, the operations performed by communications manager 912 can be implemented in program instructions configured to run on hardware, such as a processor unit. When firmware is used, the operations performed by communications manager 912 can be implemented in program instructions and data and stored in persistent memory to run on a processor unit.

When hardware is employed, the hardware may include circuits that operate to perform the operations in communications manager 912.

The circuits used to implement communications manager 912 can take other forms in addition to or in place of a processor unit.

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In the illustrative examples, the hardware used to implement communications manager **912** can take a form selected from at least one of a circuit system, an integrated circuit, an application specific integrated circuit (ASIC), a programmable logic device, or some other suitable type of hardware configured to perform a number of operations. With a programmable logic device, the device can be configured to perform the number of operations. The device can be reconfigured at a later time or can be permanently configured to perform the number of operations. Programmable logic devices include, for example, a programmable logic array, a programmable array logic, a field programmable logic array, a field programmable gate array, and other suitable hardware devices. Additionally, the processes can be implemented in organic components integrated with inorganic components and can be comprised entirely of organic components excluding a human being. For example, the processes can be implemented as circuits in organic semiconductors.

Computer system **910** is a physical hardware system and includes one or more data processing systems. In this illustrative example, the data processing systems are hardware machines that can be configured to perform a sequence of operations. These operations can be performed in response to receiving an input in generating and output based on performing the operations. This output can be data in the form of values, commands, or other types of data. When more than one data processing system is present in computer system **910**, those data processing systems are in communication with each other using a communications medium. The communications medium may be a network. The data processing systems may be selected from at least one of a computer, a server computer, a tablet, or some other suitable data processing system.

As depicted, computer system **910** includes a number of processor units **914** that are capable of executing program instructions **916** implementing processes in the illustrative examples. In other words, program instructions **916** are computer readable program instructions.

As used herein, a processor unit in the number of processor units **914** is a hardware device and is comprised of hardware circuits such as those on an integrated circuit that respond and process instructions and program code that operate a computer. When the number of processor units **914** executes program instructions **916** for a process, the number of processor units **914** can be one or more processor units that are on the same computer or on different computers. In other words, the process can be distributed between processor units **914** on the same or different computers in a computer system **910**.

Further, the number of processor units **914** can be of the same type or different type of processor units. For example, a number of processor units **914** can be selected from at least one of a single core processor, a dual-core processor, a multi-processor core, a general-purpose central processing unit (CPU), a graphics processing unit (GPU), a digital signal processor (DSP), or some other type of processor unit.

As depicted, communications system **900** can also include signal transmission system **918**. In other examples, signal transmission system **918** can be considered a separate component controlled by communications system **900**.

In this depicted example, signal transmission system **918** is a hardware system that can transmit noise signals **904**. The operation of signal transmission system **918** can be controlled by communications manager **912**.

In this illustrative example, noise signals **904** are received by receiver **920**. Receiver **920** is also depicted as part of

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communication system **900**. In yet other illustrative examples, receiver **920** may be a separate component from communications system **900**.

Receiver **920** is a hardware system and can include processes implemented in hardware or software that decode data **902** that is encoded in pulses **922** of noise signals **904**.

In this illustrative example, communications manager **912** identifies data **902** for transmission. In response to identifying data **902**, communications manager **912** transmits pulses **922** of noise signals **904** encoding data **902**. In one illustrative example, data **902** can be encoded in pulses **922** of noise signals **904** using at least one of a timing of the pulses **922**, an amplitude of the pulses **922**, duration of pulses **922**, or other characteristic for pulses **922**. In this manner, communications manager **912** can perform pulse noise modulation through the modulation of noise signals **904** to encode data **902**.

For example, communications manager **912** can control the operation of signal transmission system **918** to perform pulse modulation **924**. With pulse modulation **924**, pulses **922** can encode data **902** through the timing of pulses **922** which are noise pulses or pulses of noise in this example.

For example, the presence of a noise pulse or pulse of noise can be considered a "1" and the absence of a noise pulse or pulse of noise can be considered a "0" which can be selected in time to encode data **902**. The timing of the presence or absence of pulses **922** of noise can occur using various time periods.

For example, the timing can be based on whether a noise pulse or pulse of noise is present or absent at each period of time. The period of time can be, for example, a microsecond, a millisecond, two milliseconds, or some other period of time during which a pulse is absent or present for encoding data **902** in pulses **922** of noise.

With reference next to FIG. **10**, an illustration of a transmitter is depicted in accordance with an illustrative embodiment. In the illustrative examples, the same reference numeral may be used in more than one figure. This reuse of a reference numeral in different figures represents the same element in the different figures. In this illustrative example, examples of components that can be used to implement signal transmission system **918** in FIG. **9** are depicted.

As depicted in this illustrative example, signal transmission system **918** can include a number of different components that can be controlled to transmit noise signals **904**. More specifically, these components can be controlled to generate pulses **922** of noise signals **904**. These components can include at least one of laser generation system **1000** or radio frequency transmitter **1002**.

In this illustrative example, laser generation system **1000** is a hardware system that emits a set of laser beams **1004**. Communications manager **912** can control the emission of the set of laser beams **1004** from laser generation system **1000** to cause optical breakdowns **1005**.

In this example, optical breakdowns **1005** result in the generation of noise signals **904** in the form of radio frequency noise signals **1006**. In this example, pulses **922** of radio frequency noise signals **1006** can be generated based on the timing of optical breakdowns **1005**. In this illustrative example, each optical breakdown in optical breakdowns **1005** can be a pulse in pulses **922** of radio frequency noise signals **1006**.

In this example, radio frequency transmitter **1002** is a hardware system and can transmit pulses **922** of noise signals **904** in the form of radio frequency noise signals **1006**. For example, radio frequency transmitter **1002** can transmit pulses **922** of noise signals **904** in the form of radio

frequency noise signals **1006** transmitted from a physical hardware antenna instead of using lasers and optical breakdowns to produce the radio frequency noise signals **1006**.

Turning next to FIG. **11**, an illustration of a block diagram of a radio frequency transmitter is depicted in accordance with an illustrative embodiment. This figure illustrates example components that can be used to implement radio frequency transmitter **1002**. As depicted in this example, radio frequency transmitter **1002** comprises electrical noise generator **1100**, modulator **1102**, and transmitter **1104**.

As depicted, electrical noise generator **1100** generates carrier noise signal **1106**. Electrical noise generator **1100** is connected to modulator **1102** and sends carrier noise signal **1106** to modulator **1102**.

As depicted, modulator **1102** receives data **1108** that is to be transmitted. In this example, modulator **1102** modulates carrier noise signal **1106** to create pulsed carrier noise signal **1110** that encodes data **1108**. This data is encoded in pulses in pulsed carrier noise signal **1110**. In this example the modulation occurs by modulator **1102** turning carrier noise signal **1106** on and off to form pulsed carrier noise signal **1110**.

Transmitter **1104** transmits pulsed carrier noise signal **1110** as pulses **1112** of radio frequency noise signals **1114**. In this example, transmitter **1104** includes a physical antenna that is used to transmit pulses **1112** of radio frequency noise signals **1114**. In other illustrative examples, the antenna can be a separate component from the hardware used to generate radio frequency noise signals **1114**.

Turning next to FIG. **12**, an illustration of a block diagram of a receiver is depicted in accordance with an illustrative embodiment. An example of components that can be used to implement receiver **920** are shown in this figure. As depicted, receiver **920** is a hardware system. As depicted, receiver **920** comprises broadband radio frequency receiver **1200**, frequency selector **1202**, and clipper circuit **1204**.

In this illustrative example, broadband radio frequency receiver **1200** receives radio frequency noise signals **1206**. Broadband radio frequency receiver **1200** is connected to frequency selector **1202** and sends the received signals to frequency selector **1202**.

Frequency selector **1202** outputs a set of voltage signals **1208** from the frequencies selected in radio frequency noise signal **1206**. In this illustrative example, the selection of frequencies by frequency selector **1202** can be performed using at least one of a bandpass filter, a band-reject filter, an envelope follower, an envelope detector, a low-pass filter, a rectified low pass filter, multiple bandpass filters tuned to different frequencies, or some other suitable type of circuit.

Frequency selector **1202** is connected to clipper circuit **1204**. Voltage signal **1208** is received by clipper circuit **1204**, which shapes voltage signal **1208**. In this illustrative example, clipper circuit **1204** prevents voltage signal **1208** from exceeding a selected voltage level. Clipper circuit **1204** outputs data signal **1210**. In this example, data signal **1210** is in an analog or digital signal and contains pulses that can be used re-create the data transmitted in radio frequency noise signals **1206**.

Thus, one or more illustrative examples enable communicating data using noise carrier signals. In one illustrative example, these noise carrier signals or carrier noise signals can be modulated to encode data. The modulation can be pulse noise modulation or pulse code noise modulation in which a noise signal is transmitted in pulses. The timing of the pulses selected encodes data in these pulses of noise signals.

In this illustrative example, the modulation and demodulation of these pulses of noise signals do not depend on a single frequency or periodic waveform as the basis for the carrier wave as compared to current techniques that use a sinusoidal, periodic, or predictable carrier. As result, increased security can be present and interference with the sinusoidal carriers can be reduced.

In one illustrative example, the pulse code noise modulation or pulse noise modulation can be a broadband noise radio frequency carrier signal encoding the data. The generation of the pulses of radio frequency noise signals can be performed using a laser generation system that generates radio frequency signals through optical breakdowns. In another example, the generation of the radio frequency noise signals can be performed using a physical electromagnetic receipt transmitter having a physical antenna.

The illustration of communications environment **901** and the different components in FIGS. **9-12** is not meant to imply physical or architectural limitations to the manner in which an illustrative embodiment may be implemented. Other components in addition to or in place of the ones illustrated may be used. Some components may be unnecessary. Also, the blocks are presented to illustrate some functional components. One or more of these blocks may be combined, divided, or combined and divided into different blocks when implemented in an illustrative embodiment.

For example, although communications manager **912** is shown as being implemented using program instructions **916** run on a number of processor units **914** in computer system **910**, communications manager **912** can be implemented in other hardware instead of or in addition to the number of processor units **914**. For example, computer system **910** may use other hardware in addition to or in place of the number of processor units **914**.

For example, other types of hardware circuits capable of performing the operations for communications manager **912** can be used. This other hardware can be at least one of a circuit system, an integrated circuit, an application specific integrated circuit (ASIC), a programmable logic device, or some other suitable type of hardware configured to perform a number of operations.

As another example, the illustration of laser generation system **1000** and radio frequency transmitter **1002** are provided as examples of some implementations of components that can transmit pulses **922** of noise signals **904**. As another example, radio frequency transmitter **1002** that generates electrical noise in electrical noise generator **1100** as carrier signal noise **1106** to be modulated and transmitted as radio frequency noise signals **1114** in noise pulses **1112** can be transmitted on any type of physical, hardware antenna, or both. Examples of antenna types include, for example, whip antennas, dipole antennas, microwave antennas, metamaterial antennas, directional antennas, omnidirectional antennas, and any other type of physical antenna.

Turning next to FIG. **13**, an illustration of a communications system for transmitting and receiving electromagnetic noise signals is depicted in accordance with an illustrative embodiment. In this illustrative example, communications system **1300** can transmit or receive electromagnetic noise signals encoding data using electromagnetic noise signals. Examples of electromagnetic noise signals may include electromagnetic ranges of ULF (Ultra Low Frequency), VLF (Very Low Frequency), 20 KHz to 300 KHz, HF (High Frequency), UHF (Ultra High Frequency), millimeter wave and microwave ranges, EHF (Extremely High Frequencies) up through Gigahertz frequencies and above Terahertz frequencies, and including the optical spectrum. Examples of

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applications or uses include noise carrier communications for modulation of audio, voice, and video communications, as well as noise-based radar, noise-based precision navigation and timing such as noise-based global positioning systems, noise-based spread spectrum using frequency bands of noise instead of sinusoidal-based carrier spread spectrum, noise-based frequency band-hopping using frequency-hopping of frequency bands of noise instead of using sinusoidal or periodic-based carrier frequency-hopping, as well as Signals Intelligence (SI) waveforms such as Low Probability of Intercept/Low Probability of Detect (LPI/LPD) and other clandestine signaling where detection and interception of messages using noise carriers will be difficult.

As depicted, modulator 1302 receives data signal 1304 and carrier noise signal 1306. In this illustrative example, carrier noise signal 1306 can be generated by an electrical noise generator.

Modulator 1302 modulates carrier noise signal 1306 to generate modulated signal 1308. In this example, modulated signal 1308 comprises pulses of carrier noise signal 1306. For example, modulated signal 1308 can be generated by turning modulator 1302 on and off to send pulses of carrier noise signal 1306 to transmitter 1310 for transmission as modulated signal 1308. The generation of the pulses is based on the data in data signal 1304. In this manner, the data in data signal 1304 can be encoded in modulated signal 1308.

Transmitter 1310 transmits modulated signal 1308 to receiver 1312. In one illustrative example, receiver 1312 can be a broadband radio frequency receiver when modulated signal 1308 is a radio frequency signal. When other types of signals are used, receiver 1312 is selected to detect the signals transmitted by transmitter 1310.

Modulated signal 1308 detected by receiver 1312 is sent to demodulator 1314. In this example, demodulator 1314 demodulates modulated signal 1308 using carrier noise signal 1320 to generate data signal 1318, which contains the same data in data signal 1304 in this depicted example.

As depicted, the demodulation of modulated signal 1308 is performed using carrier noise signal 1320. In this illustrative example, carrier noise signal 1306 is not predictable as compared to current techniques using sinusoidal waveforms for carrier signals.

As depicted, carrier noise signal 1320 can be obtained by demodulator 1314 in the form of unmodulated carrier noise signal 1322 being transmitted to demodulator 1314. In this manner, carrier noise signal 1320 used to demodulate modulated signal 1308 can be the same carrier signal as carrier noise signal 1306. Unmodulated carrier noise signal 1322 can be an in-band or out-of-band copy of carrier noise signal 1306.

With reference now to FIG. 14, an illustration of a block diagram of a communications system for transmitting and receiving electromagnetic noise signals is depicted in accordance with an illustrative embodiment. Communications system 1400 can transmit or receive electromagnetic noise signals encoding data using electromagnetic noise signals.

In this illustrative example, communications system 1400 can transmit or receive electromagnetic noise signals encoding data using electromagnetic noise signals.

As depicted, modulator 1402 receives data signal 1404 and carrier noise signal 1406. In this illustrative example, carrier noise signal 1406 can be generated by an electrical noise generator.

Modulator 1402 modulates carrier noise signal 1406 to generate modulated signal 1408. In this example, modulator 1402 can be an on/off modulator. As an on/off modulator,

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modulator 1402 sends carrier noise signal 1406 transmitter 1410 for transmission when modulator 1402 is turned on and does not send carrier noise signal 1406 to transmitter 1410 when modulator 1402 is turned off. As result, modulated signal 1408 comprises pulses of carrier noise signal 1406. These pulses are generated to encode data signal 1404. In other words, the timing of these pulses can be generated to encode the data. For example, the timing in these depicted examples can be time for pulses to perform pulse code noise modulation or pulse noise modulation.

For example, modulated signal 1408 can be generated by turning modulator 1402 on and off to send pulses of carrier noise signal 1406 to transmitter 1410 for transmission as modulated signal 1408.

Transmitter 1410 transmits modulated signal 1408 to receiver 1412. In one illustrative example, receiver 1412 can be a broadband radio frequency receiver when modulated signal 1408 is a radio frequency signal. When other types of signals are used, receiver 1412 is selected to detect the signals transmitted by transmitter 1410.

In this example, modulated signal 1408 detected by receiver 1412 is sent to envelope follower 1414. As depicted, envelope follower 1414 can also be referred to as an envelope detector. Envelope follower 1414 can detect amplitude variations in modulated signal 1408 and create a signal having a shape that resembles those variations. This example, modulated signal 1408 contains pulses of noise. As a result, envelope follower 1414 can generate a signal with the shape of the noise pulses to form data signal 1416. Envelope follower 1414 can be selected from at least one of a low pass filter, a bandpass filter, an envelope detector, a peak detector, or a diode detector that follows and outputs the overall shape of at least one of the amplitudes or pulses as currently used.

The illustrative examples of communication systems in FIG. 13 and in FIG. 14 are presented as examples of some implementations for communications system 900 in FIG. 9. These illustrations are not meant to limit the manner in which other illustrative examples can be implemented. For example, a clipper circuit as is known in the art can be placed after envelope follower 1414 in FIG. 14 to convert rough envelopes of pulses into square wave pulses.

Turning now to FIG. 15A, an illustration of a data flow of signals transmitting data using modulated noise signals is depicted in accordance with an illustrative embodiment. In this illustrative example, data signal 1500 is an example of signals in communications system 1400 in FIG. 14.

In this illustrative example, data signal 1500 is used to modulate carrier noise signal 1502. Data signal 1500 is an example of data signal 1404 and carrier noise signal 1502 is an example of carrier noise signal 1406 in FIG. 14.

The modulation of carrier noise signal 1502 forms modulated signal 1504, which encodes the data in data signal 1500. Modulated signal 1504 is an example of modulated signal 1408 in FIG. 14. As depicted in this example, modulated signal 1504 is a modulated noise signal comprising pulses of carrier noise signal 1502.

Received signal 1506 is an example of the signal received by a receiver. As depicted, received signal 1506 also includes noise 1508 in addition to the pulses of carrier noise signal 1502 in modulated signal 1504. In this example, noise 1508 is background noise or other noise in addition to the pulses in the carrier noise in modulated signal 1504.

As depicted, received signal 1506 can be processed and decoded using a component such as envelope follower 1414 in FIG. 14. Other components such as a bandpass filter, low-pass filter, band reject filter, clipper circuit, or other

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circuits can be used to generate data signal **1510**. In this example, data signal **1510** is the same as or close enough to data signal **1500** such that the same data used to generate data signal **1500** can be obtained from data signal **1510**.

As discussed previously, the set of characteristics for noise signals can be selected from at least one of a timing, an amplitude, a frequency band, a relative phase, or other characteristics for carrier noise signals. For pulse noise modulation the carrier noise may be of different frequency characteristics that the transmitter and receiver will share. For pulse noise modulation the carrier noise signals will vary in amplitude, duration, and timing to modulate the message signal. For reception of these pulse noise modulated signals the receivers in FIG. **12** and FIG. **14** use various types of techniques to receive and demodulate the original data signal.

Turning now to FIG. **15B**, an illustration of an envelope follower circuit using a diode detector with a low pass filter in accordance with an illustrative embodiment. In this figure, an illustration of a simple circuit for envelope follower **1414** is shown. In this illustrative example, envelope follower **1414** is comprised of a diode **1512** to rectify the input signal, capacitor **1514** to provide a low pass filter to smooth out the noisy rectified signal and produce a lower frequency envelope. An optional resistor **1516** or inductive coil may be provided to affect the tuning or resonance of the circuit. Here, received signal **1506** of carrier pulses of noise is inputted to the envelope follower **1414** circuit. As received signal **1506** travels through diode **1512** the diode acts as a rectifier and converts the AC noise signal into a DC noise signal **1518** as shown by the dashed arrow from DC noise signal **1518** to the output of diode **1512**. From there the rectified DC noise signal **1518** travels across capacitor **1514** which acts as a low pass filter to smooth the signal into an envelope signal **1520**. The actual envelope signal **1520** is shown by the line that follows the outline or envelope of the noise bursts from DC noise signal **1518**. The envelope signal **1520** then travels across optional resistor **1516** or coil and exits at the output as the envelope followed signal **1522**.

In this illustrative example it is clear that the envelope followed signal **1522** is beginning to look like the received data signal **1510**.

With reference to FIG. **15C**, an illustration of a clipper circuit in accordance with an illustrative embodiment. In this figure, clipper circuit **1204** is also referred to as a slicer or amplitude selector. In this illustrative example, clipper circuit **1204** is comprised of optional input resistor **1524**, and a bidirectional clipping circuit comprised of diode **D1 1526**, bias voltage **1528**, diode **D2 1530**, and bias voltage **1532**. This and many other known methods of clipping can be used. Single directional clipping may be used as well as bidirectional or any other type of clipping circuit.

In this illustrative example, envelope followed signal **1522** from FIG. **15B** has been amplified to be a stronger signal and is inputted into clipper circuit **1204**. Envelope followed signal **1522** travels through optional input resistor **1524** which may be an impedance matching circuit.

This signal then travels across one or more illustrative diodes **D1 1526** and **D2 1530**. Various types of diodes may be used. A single diode may be used, or a transistor circuit may be used with the purpose of clipping off the top of envelope followed signal **1522** such that top part of signal **1534** is clipped off and bottom part of signal **1536** remains. The level at which top part of the signal **1534** is clipped off is determined by the diodes **D1 1526** and **D2 1530** as well as by the bias voltages **1528** and **1532**.

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Thus, bottom part of signal **1536** remaining is outputted at the output. This bottom part of the signal **1536** can be transferred through another stage of clipping until it becomes output data signal **1510** which is extremely similar to the original data signal **1500**.

As can be seen in this illustrative example, the pulses of carrier noise signal **1502** encode data in data signal **1500**. In other words, the timing in generating pulses of carrier noise signal **1502** is used to encode the data.

Thus, the different illustrative examples use pulse modulation of a noise signal that can be generated using a laser generator or a transmitter. With a laser generator, optical breakdowns are used to create the pulses of noise signals. With a physical transmitter, an electronic noise source generates a carrier noise signal that is modulated to create pulses of the carrier noise signal based on the data to be transmitted. These pulses of the carrier noise signals form the pulses of noise signal encoding data that can be transmitted using a physical antenna.

In this illustrative example, FIGS. **16-27** are flowcharts illustrating operations that can be performed to generate radio frequency noise signals encoding data in which a physical antenna is unnecessary.

With reference first to FIG. **16**, a flowchart of a process for transmitting data is depicted in accordance with an illustrative embodiment. The process in FIG. **16** can be implemented in hardware, software, or both. When implemented in software, the process can take the form of program instructions that is run by one of more processor units located in one or more hardware devices in one or more computer systems. For example, the process can be implemented in communications manager **212** in computer system **210** in FIG. **2**.

The process begins by identifying data for transmission (operation **1600**). The process controls an emission of a set of laser beams to cause optical breakdowns generating radio frequency noise signals encoding the data (operation **1602**). The process terminates thereafter.

In operation **1602**, the emission of the set of laser beams can be controlled in number of different ways. For example, the laser beams can be emitted continuously or pulsed. Further, direction at which the laser beams are directed can also be changed. For example, the set of laser beams can be directed towards a set of optical breakdown points. The optical breakdown points can be selected from at least one of an intersection point or focal point. These optical breakdown points are locations where optical breakdowns occur. These optical breakdowns are locations where plasma is generated that generates the radio frequency noise signals.

The manner in which the optical breakdowns occur can be used to encode the data in the radio frequency noise signals. For example, the timing of the occurrence of optical breakdowns generate time pulses used to encode data. In this manner, different types of data encoding such as pulse noise modulation can be used to encode data based on when radio frequency noise signals are generated.

As another example, the set of laser beams can be moved or swept such that the optical breakdowns occur in different locations resulting in the frequency of a phase change in the optical breakdowns that can be used to encode data. As another example, the power of the laser beams can be changed to change the amplitude of the radio frequency noise signals two encode data. In this manner, different types of data encoding such as pulse noise modulation can be used to encode data based on when radio frequency noise signals are generated.

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Turning next to FIG. 17, an illustration of a flowchart for controlling the emission of laser beams is depicted in points with an illustrative embodiment. The process illustrated in FIG. 17 is an example of one implementation for operation 1602 in FIG. 16.

The process controls a power of a laser beam in the set of laser beams to reach an optical breakdown level at a focal point to cause the optical breakdowns that generate the radio frequency noise signals encoding the data (operation 1700). The process terminates thereafter.

With reference next to FIG. 18, an illustration of a flowchart for controlling the emission of laser beams is depicted in points with an illustrative embodiment. The process illustrated in FIG. 18 is another example of an implementation for operation 1602 in FIG. 16.

The process controls a power of a laser beam in the set of laser beams to reach an optical breakdown level at a focal point to cause the optical breakdowns that generate the radio frequency noise signals encoding the data (operation 1800). The process terminates thereafter.

In FIG. 19, an illustration of a flowchart for controlling the emission of laser beams is depicted in points with an illustrative embodiment. The process illustrated in FIG. 19 is yet another example of an implementation for operation 1602 in FIG. 16.

The process controls emission of the set of laser beams to intersect an intersection point such that a power of the set of the laser beams at the intersection point causes the optical breakdowns that generate the radio frequency noise signals encoding the data (operation 1900). The process terminates thereafter.

Turning next to FIG. 20, a flowchart of a process for transmitting data is depicted in accordance with an illustrative embodiment. The process in FIG. 20 can be implemented in hardware, software, or both. When implemented in software, the process can take the form of program instructions that is run by one of more processor units located in one or more hardware devices in one or more computer systems. For example, the process can be implemented in communications manager 212 in computer system 210 in FIG. 2.

The process begins by identifying the data for transmission using radio frequency noise signals (operation 2000). The process controls an emission of laser beams at a set of optical breakdown points to cause optical breakdowns that generate the radio frequency noise signals encoding the data (operation 2002). The process terminates thereafter.

In this example, the set of optical breakdown points can be at different locations when more than one optical breakdown point is present in the set of optical breakdown points. In one example, radio frequency transmissions can be transmitted from multiple locations when the set of optical breakdowns caused by the set of lasers being directed at more than one optical breakdown point.

In FIG. 21, an illustration of a flowchart for controlling the emission of laser beams is depicted in points with an illustrative embodiment. The process illustrated in FIG. 21 is yet another example of an implementation for operation 2002 in FIG. 20.

The process begins by emitting a first set of the laser beams continuously at the set of optical breakdown points (operation 2100). The process pulses a second set of the laser beams at the set of optical breakdown points to cause the optical breakdowns that generate the radio frequency noise signals encoding the data (operation 2102). The process terminates thereafter.

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In operation 2102, the pulsing can occur by turning the second set of laser beams on and off. In other examples, the pulsing can provide increasing decreasing the power to the second set of laser beams. In this example, the optical breakdowns occur in response to sufficient power in the laser beams at the set of optical breakdown points. In this example, the pulsing can control the timing of when radio frequency noise signals are transmitted.

Further in operation 2102, a power of a laser beam at the optical breakdown point can be controlled at by at least one of a shutter, a lens, a deformable lens, a microelectromechanical systems mirror, an attenuator, a controlling optics, an optical filter, an amplitude modulator in a laser beam generator, or other suitable components.

With reference now to FIG. 22, an illustration of a flowchart for controlling the emission of the laser beams is depicted in points with an illustrative embodiment. The process illustrated in FIG. 21 is yet another example of an implementation for operation 2002 in FIG. 20.

The process emits the laser beams at the set of optical breakdown points causing the optical breakdowns that generate the radio frequency noise signals encoding the data (operation 2200). The process terminates thereafter.

Turning next to FIG. 23, an illustration of a flowchart for controlling the emission of laser beams is depicted in points with an illustrative embodiment. The process illustrated in FIG. 23 is example of an implementation for operation 2200 in FIG. 22.

The process begins by emitting the laser beams at a selected optical breakdown point in the set of optical breakdown points (operation 2300). The process selects a new optical breakdown point in the set of optical breakdown points as the selected optical breakdown point in response to a set of optical breakdowns occurring at the selected optical breakdown point (operation 2302).

The process repeats emitting the set of laser beams and selecting a new optical breakdown point while generating the radio frequency noise signals encoding the data (operation 2304) the process terminates thereafter. In operation 2304, the process repeats operations 2300 and operation 2302 any number of times while transmitting the radio frequency noise signals. Operation at 2304 enables transmitting the radio frequency signals from different locations through the selection of different optical breakdown points. As result, identifying the origination of the radio frequency signals can be made more difficult.

With reference next to FIG. 24, an illustration of a flowchart for controlling the emission of laser beams is depicted in points with an illustrative embodiment. The process illustrated in FIG. 24 is another example of an implementation for operation 2002 in FIG. 20.

The process begins by emitting a subset of the laser beams at the set of optical breakdown points to cause the optical breakdowns that generate the radio frequency noise signals encoding the data (operation 2400). The process selects a new subset of laser beams as the subset of the laser beams (operation 2402).

The process repeats emitting the subset of laser beams and selecting a new subset of laser beams while transmitting the radio frequency noise signals encoding the data (operation 2404). The process terminates thereafter. By using different subsets of the laser beams, identifying a location from which the laser beams originate can be made more difficult when the laser beams are emitted from different locations.

In FIG. 25, an illustration of a flowchart for controlling the emission of laser beams is depicted in points with an

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illustrative embodiment. The process illustrated in FIG. 25 is an example of an implementation for operation 2002 in FIG. 20.

The process emits the set of laser beams from different locations at an optical breakdown point, wherein a portion of the set of laser beams intersect at the optical breakdown point such that a power from the portion of the laser beams is sufficient to cause the optical breakdowns at the intersection point that generate the radio frequency noise signals encoding the data (operation 2500). The process terminates thereafter.

With reference to FIG. 26, an illustration of a flowchart for controlling the emission of laser beams is depicted in points with an illustrative embodiment. The process illustrated in FIG. 26 is an example of an implementation for operation 2002 in FIG. 20.

The process emits the laser beams at an optical breakdown point (operation 2600). The process terminates thereafter. In operation 2600, the optical breakdowns occur in response to all of the laser beams intersecting at the optical breakdown point.

With reference now to FIG. 27, an illustration a flowchart for controlling laser beams is depicted in accordance with an illustrative embodiment. The process illustrated in this figure is an example of an additional operation that can be performed with the operations in FIG. 20.

The process changes a set of laser beam parameters for the laser beams to encode the data into the radio frequency noise signals (operation 2700). The process terminates thereafter. In operation of 2700, changing the set of laser beam parameters changes a set of radio frequency characteristics for the radio frequency noise signals. The set of radio frequency characteristics is selected from at least one of a timing, an optical breakdown point, an amplitude of the radio frequency noise signals, or other characteristics of the radio frequency noise signals.

In this illustrative example, FIGS. 28-35 are flowcharts illustrating operations that can be performed to encode data in noise signals. Turning first to FIG. 28, an illustration of a flowchart for communicating data is depicted in accordance with an illustrative embodiment. The process in FIG. 28 can be implemented in hardware, software, or both. When implemented in software, the process can take the form of program instructions that is run by one of more processor units located in one or more hardware devices in one or more computer systems. For example, the process can be implemented in communications manager 912 in computer system 910 in FIG. 9.

The process begins by identifying data for transmission (operation 2800). The process transmits pulses of noise signals encoding the data (operation 2802). The process terminates thereafter. The pulses of noise signals can be selected from at least one of electromagnetic frequency signals, radio frequency signals, microwave frequency signals, audio frequency signals, ultrasonic frequency signals, ultra-low frequency signals, very low frequency signals, underwater frequency signals, or optical frequency signals.

In operation 2802, the pulses of radio frequency noise signals can be transmitted in a number of different ways. For example, these pulses of noise signals can be radio frequency noise signals transmitted from a physical antenna. In another illustrative example, the pulses of noise signals can be transmitted using optical breakdowns generated by laser beams. The optical breakdowns can be controlled to generate pulses of noise signals in the form of radio frequency noise signals.

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The noise signals can be generated using at least one of a laser generation system that emits lasers to cause optical breakdown that generates the noise signal or an electric noise generator. The noise in the noise signal can be selected from at least one of nondeterministic noise or pseudo random noise.

Turning to FIG. 29, an illustration of a flowchart for transmitting pulses of noise signals is depicted in accordance with an illustrative embodiment. This flowchart is an example of an implementation for operation 2802 in FIG. 28. In this example, the pulses of noise signals can be pulses of radio frequency noise signals.

The process controls emission of a set of laser beams from a laser beam generator to cause optical breakdowns that generate the pulses of the radio frequency noise signals that encode the data (operation 2900). The process terminates thereafter.

With reference next to FIG. 30, another illustration of a flowchart for transmitting pulses of noise signals is depicted in accordance with an illustrative embodiment. This flowchart is an example of an implementation for operation 2802 in FIG. 28.

The process begins by generating a carrier radio frequency noise signal (operation 3000). The process modulates the carrier noise signal to form the pulses of the noise signals (operation 3002). In operation 3002, the pulses encode the data.

The process transmits the pulses of noise signals (operation 3004). The process terminates thereafter.

Turning now to FIG. 31, an illustration of a flowchart for communicating data is depicted in accordance with an illustrative embodiment. The process in FIG. 31 can be implemented in hardware, software, or both. When implemented in software, the process can take the form of program instructions that is run by one of more processor units located in one or more hardware devices in one or more computer systems. For example, the process can be implemented in communications manager 912 in computer system 910 in FIG. 9.

The process begins by identifying data for transmission (operation 3100). The process controls emission of a set of laser beams to cause optical breakdown that generate pulses of radio frequency noise signals (operation 3102). The process terminates thereafter. In operation 3100, the data can be encoded in the pulses of the radio frequency noise signals.

With reference to FIG. 32, an illustration of a flowchart of a process for controlling the mission of a set of laser beams is depicted in accordance with an illustrative embodiment. The process in this flowchart is an example of an implementation for operation 3102 in FIG. 31.

The process controls a power of a laser beam in the set of laser beams to reach an optical breakdown level at a focal point to cause the optical breakdowns that generate the pulses of radio frequency noise signals encoding the data (operation 3200). The process terminates thereafter.

Turning next to FIG. 33, an illustration of a flowchart of a process for controlling the emission of a set of laser beams is depicted in accordance with an illustrative embodiment. The process in this flowchart is an example of an implementation for operation 3102 in FIG. 31.

The process controls the controlling emission of the set of laser beams to intersect an intersection point such that the power of the set of the laser beams at the intersection point causes the optical breakdowns that generate the pulses of the radio frequency noise signals encoding the data (operation 3300). The process terminates thereafter.

In FIG. 34, an illustration of a flowchart for communicating data is depicted in accordance with an illustrative embodiment. The process in FIG. 34 can be implemented in hardware, software, or both. When implemented in software, the process can take the form of program instructions that is run by one of more processor units located in one or more hardware devices in one or more computer systems. For example, the process can be implemented in communications manager 912 in computer system 910 in FIG. 9.

The process begins by receiving pulses of noise signals (operation 3400). In operation 3400, data is encoded in the pulses of noise signals.

The process decodes the data encoded in the pulses of the noise signals using a set of characteristics of the pulses of the noise signals (operation 3402). The process terminates thereafter. In operation 3402, the set of characteristics comprises at least one of a timing of the pulses of noise, an amplitude of the pulses of noise, a duration of the pulses of noise, or some other characteristic.

With reference now to FIG. 35, an illustration of a flowchart for decoding data is depicted in accordance with an illustrative embodiment. The process depicted in this flowchart is an example of an implementation for operation 3402 in FIG. 34.

The process begins by receiving signals in a frequency range that includes the pulses of the noise signals encoding the data (operation 3500). In operation 3500, the signals in frequency range can be received using at least one of a bandpass filter, a notch filter, a band reject filters

The process identifies the pulses of the noise signals in the frequency range (operation 3502). The process terminates thereafter. In operation 3502, the pulses of the noise signals in the frequency range can be identified using an envelope detector.

The flowcharts and block diagrams in the different depicted embodiments illustrate the architecture, functionality, and operation of some possible implementations of apparatuses and methods in an illustrative embodiment. In this regard, each block in the flowcharts or block diagrams can represent at least one of a module, a segment, a function, or a portion of an operation or step. For example, one or more of the blocks can be implemented as program instructions, hardware, or a combination of the program instructions and hardware. When implemented in hardware, the hardware can, for example, take the form of integrated circuits that are manufactured or configured to perform one or more operations in the flowcharts or block diagrams. When implemented as a combination of program instructions and hardware, the implementation may take the form of firmware. Each block in the flowcharts or the block diagrams can be implemented using special purpose hardware systems that perform the different operations or combinations of special purpose hardware and program instructions run by the special purpose hardware.

In some alternative implementations of an illustrative embodiment, the function or functions noted in the blocks may occur out of the order noted in the figures. For example, in some cases, two blocks shown in succession may be performed substantially concurrently, or the blocks may sometimes be performed in the reverse order, depending upon the functionality involved. Also, other blocks may be added in addition to the illustrated blocks in a flowchart or block diagram.

Thus, the illustrative examples provide a method, apparatus, and system for transmitting radio frequency signals using a transmission system in which a physical antenna is absent. Optical breakdowns are generated by laser beams in

which the optical breakdowns create plasma. The plasma results in radio frequency noise signals. The optical breakdowns can be controlled to encode data in the radio frequency noise signals. The locations of these optical breakdowns are radio frequency source emitters in the depicted examples.

Further, these radio frequency source emitters can be moved to different locations by repositioning the laser beams such that the laser beams point at different optical breakdown points. Attacks at these locations are in essence attacks at the optical breakdown points where the plasma is generated.

As a result, kinetic attacks against these locations are useless because no physical infrastructure is present at the locations. Further, the laser modulation sources are remote from the locations of these radio frequency source emitters. These optical breakdowns can occur at a location that is remote from the laser source.

Further, the illustrative examples can encode data using noise signals. The use of noise signals is in contrast to the use of sinusoidal signals as a carrier signal to encode data. With the encoding of data in pulses of noise signals, issues with detection and interference in transmitting data encoded using sinusoidal carriers can be reduced.

The description of the different illustrative embodiments has been presented for purposes of illustration and description and is not intended to be exhaustive or limited to the embodiments in the form disclosed. The different illustrative examples describe components that perform actions or operations. In an illustrative embodiment, a component can be configured to perform the action or operation described. For example, the component can have a configuration or design for a structure that provides the component an ability to perform the action or operation that is described in the illustrative examples as being performed by the component. Further, to the extent that terms “includes”, “including”, “has”, “contains”, and variants thereof are used herein, such terms are intended to be inclusive in a manner similar to the term “comprises” as an open transition word without precluding any additional or other elements.

Many modifications and variations will be apparent to those of ordinary skill in the art. Further, different illustrative embodiments may provide different features as compared to other desirable embodiments. The embodiment or embodiments selected are chosen and described in order to best explain the principles of the embodiments, the practical application, and to enable others of ordinary skill in the art to understand the disclosure for various embodiments with various modifications as are suited to the particular use contemplated.

What is claimed is:

1. A radio frequency communications system comprising:
 - a computer system;
 - a communications manager in the computer system, wherein the communications manager is configured to:
 - identify data for transmission using radio frequency noise signals; and
 - control an emission of a set of laser beams at a set of optical breakdown points to cause optical breakdowns that generate the radio frequency noise signals encoding the data,
 - wherein in controlling the emission of the set of laser beams, the communications manager is configured to:
 - emit the set of laser beams from different locations at an optical breakdown point, wherein a portion of the set of laser beams intersect at the optical breakdown point such that a power from the portion of the set of laser

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beams is sufficient to cause the optical breakdowns at the optical breakdown point that generate the radio frequency noise signals encoding the data.

2. The radio frequency communications system of claim 1 further comprising:

a laser generation system in communication with the communications manager, wherein the communications manager controls the laser generation system to emit the set of laser beams.

3. The radio frequency communications system of claim 1, wherein in controlling the emission of the set of laser beams, the communications manager is configured to:

emit a first number of the set of laser beams continuously at the set of optical breakdown points; and

pulse a second number of the set of laser beams at the set of optical breakdown points to cause the optical breakdowns that generate the radio frequency noise signals encoding the data.

4. The radio frequency communications system of claim 3, wherein the set of laser beams are emitted from different locations.

5. The radio frequency communications system of claim 1, wherein in controlling the emission of the set of laser beams, the communications manager is configured to:

emit the set of laser beams at the set of optical breakdown points causing the optical breakdowns that generate the radio frequency noise signals encoding the data.

6. The radio frequency communications system of claim 5, wherein in emitting the laser beams at the set of optical breakdown points, the communications manager is configured to:

emit the set of laser beams at a selected optical breakdown point in the set of optical breakdown points;

select a new optical breakdown point in the set of optical breakdown points as the selected optical breakdown point; and

repeat the emitting the set of laser beams and selecting the new optical breakdown point while generating the radio frequency noise signals encoding the data.

7. The radio frequency communications system of claim 1, wherein in controlling the emission of the set of laser beams, the communications manager is configured to:

emit a subset of the set of laser beams at the set of optical breakdown points to cause the optical breakdowns that generate the radio frequency noise signals encoding the data;

select a new subset of the set of laser beams as the subset of the set of laser beams; and

repeat emitting of the subset of laser beams and selecting the new subset of the set of laser beams while transmitting the radio frequency noise signals encoding the data.

8. The radio frequency communications system of claim 6, wherein the set of laser beams are emitted from different locations.

9. The radio frequency communications system of claim 1, wherein in controlling the emission of the set of laser beams, the communications manager is configured to:

emit the set of laser beams at an optical breakdown point, wherein the optical breakdowns occur in response to all of the set of laser beams intersecting at the optical breakdown point.

10. The radio frequency communications system of claim 1, wherein a power of a laser beam at an optical breakdown point is controlled by at least one of a shutter, a lens, a deformable lens, a microelectromechanical systems mirror,

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an attenuator, a controlling optics, an optical filter, or an amplitude modulator in a laser beam generator.

11. The radio frequency communications system of claim 1, wherein the communications manager is configured to:

change a set of laser beam parameters for the set of laser beams to encode the data into the radio frequency noise signals, wherein changing the set of laser beam parameters changes a set of radio frequency characteristics for the radio frequency noise signals.

12. The radio frequency communications system of claim 11, wherein the set of radio frequency characteristics for the radio frequency noise signals is selected from at least one of a timing, an optical breakdown point, or an amplitude of the radio frequency noise signals.

13. The radio frequency communications system of claim 1, wherein the set of optical breakdown points is selected from at least one of an intersection point or a focal point.

14. A radio frequency communications system comprising:

a laser generation system configured to emit a set of laser beams;

a computer system; and

a communications manager in the computer system, wherein the communications manager is configured to:

identify data for transmission; and
control an emission of the set of laser beams using the laser generation system to cause optical breakdowns generating radio frequency noise signals encoding the data,

wherein in controlling the emission of the set of laser beams, the communications manager is configured to:
emit the set of laser beams at the set of optical breakdown points causing the optical breakdowns that generate the radio frequency noise signals encoding the data, and

wherein in emitting the laser beams at the set of optical breakdown points, the communications manager is configured to:

emit the set of laser beams at a selected optical breakdown point in the set of optical breakdown points;

select a new optical breakdown point in the set of optical breakdown points as the selected optical breakdown point; and

repeat the emitting the set of laser beams and selecting the new optical breakdown point while generating the radio frequency noise signals encoding the data.

15. A method for transmitting data, the method comprising:

identifying the data for transmission; and

controlling an emission of a set of laser beams to cause optical breakdowns generating radio frequency noise signals encoding the data,

wherein controlling the emission of the laser beams comprises:

emitting a subset of the laser beams at the set of optical breakdown points to cause the optical breakdowns that generate the radio frequency noise signals encoding the data;

selecting a new subset of the laser beams as the subset of the laser beams; and

repeating the emitting the subset of the laser beams and selecting the new subset of the laser beams while transmitting the radio frequency noise signals encoding the data.

16. The method of claim 15, wherein controlling the emission of the set of laser beams comprises:

controlling a power of a laser beam in the set of laser beams to reach an optical breakdown level at a focal

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point to cause the optical breakdowns that generate the radio frequency noise signals encoding the data.

17. The method of claim 15, wherein controlling an emission of the set of laser beams comprises:

controlling emission of the set of laser beams to intersect
an intersection point such that a power of the set of the
laser beams at the intersection point causes the optical
breakdowns that generate the radio frequency noise
signals encoding the data.

18. A method for transmitting data, the method comprising:

identifying the data for transmission using radio frequency noise signals; and

controlling an emission of laser beams at a set of optical
breakdown points to cause optical breakdowns that
generate the radio frequency noise signals encoding the
data;

wherein controlling the emission of the laser beams
comprises:

emitting a first set of the laser beams continuously at the
set of optical breakdown points; and

pulsing a second set of the laser beams at the set of optical
breakdown points to cause the optical breakdowns that
generate the radio frequency noise signals encoding the
data.

19. The method of claim 18, wherein controlling the emission of the laser beams comprises:

emitting the laser beams at the set of optical breakdown
points causing the optical breakdowns that generate the
radio frequency noise signals encoding the data.

20. The method of claim 19, wherein emitting the laser beams at the set of optical breakdown points comprises:

emitting the laser beams at a selected optical breakdown
point in the set of optical breakdown points; and

selecting a new optical breakdown point in the set of
optical breakdown points as the selected optical breakdown
point in response to a set of optical breakdowns
occurring at the selected optical breakdown point; and
repeating the emitting the laser beams and selecting new
optical breakdown points while generating the radio
frequency noise signals encoding the data.

21. The method of claim 18, wherein controlling the emission of the laser beams comprises:

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emitting a subset of the laser beams at the set of optical
breakdown points to cause the optical breakdowns that
generate the radio frequency noise signals encoding the
data;

selecting a new subset of the laser beams as the subset of
the laser beams; and

repeating the emitting the subset of the laser beams and
selecting the new subset of the laser beams while transmitting
the radio frequency noise signals encoding the data.

22. The method of claim 21, wherein the set of laser
beams are emitted from different locations.

23. The method of claim 18, wherein controlling the emission of the laser beams comprises:

emitting the laser beams from different locations at an
optical breakdown point, wherein a portion of the set of
laser beams intersect at the optical breakdown point
such that a power from the portion of the set of laser
beams is sufficient to cause the optical breakdowns at
the optical breakdown point that generate the radio
frequency noise signals encoding the data.

24. The method of claim 18, wherein controlling the emission of the laser beams comprises:

emitting the laser beams at an optical breakdown point,
wherein the optical breakdowns occur in response to all
of the laser beams intersecting at the optical breakdown
point.

25. The method of claim 18, wherein a power of a laser
beam at an optical breakdown point is controlled at by at
least one of a shutter, a lens, a deformable lens, a micro-
electromechanical systems mirror, an attenuator, a controlling
optics, an optical filter, or an amplitude modulator in a
laser beam generator.

26. The method of claim 18 further comprising:

changing a set of laser beam parameters for the laser
beams to encode the data into the radio frequency noise
signals, wherein changing the set of laser beam parameters
changes a set of radio frequency characteristics for the
radio frequency noise signals.

27. The method of claim 26, wherein the set of radio
frequency characteristics is selected from at least one of a
timing, an optical breakdown point, or an amplitude of the
radio frequency noise signals.

28. The method of claim 27, wherein the set of optical
breakdown points is selected from at least one of an intersection
point or a focal point.

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